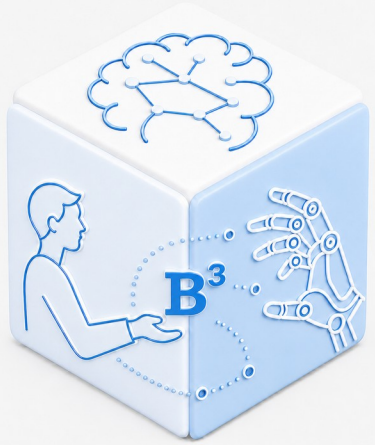
Embodied AI with Web-Scale Video Data

EN.601.439/639

Instructor: Homanga Bharadhwaj



Lecture 13

World Models

embodied-ai-with-videos.github.io

The Brain is a Prediction Machine

The brain is a dynamic prediction machine that constantly forecasts what will happen next rather than passively recording the outside world



Generative models invert hidden causes

Latent causes z generate sensory data y through p(y | z).

Perception approximates the posterior:

$p(z \mid y) \propto p(y \mid z)\, p(z)$



SAME SIGNAL → DIFFERENT UNDERLYING CAUSAL MECHANISM

Active inference closes the perception–action loop

Perception

Change internal states to reduce prediction error

Action

Change sampled sensations to fulfill predictions





predictive processing provides a unifying framework for cognitive science through the minimization of prediction error

Predictive processing in the brain



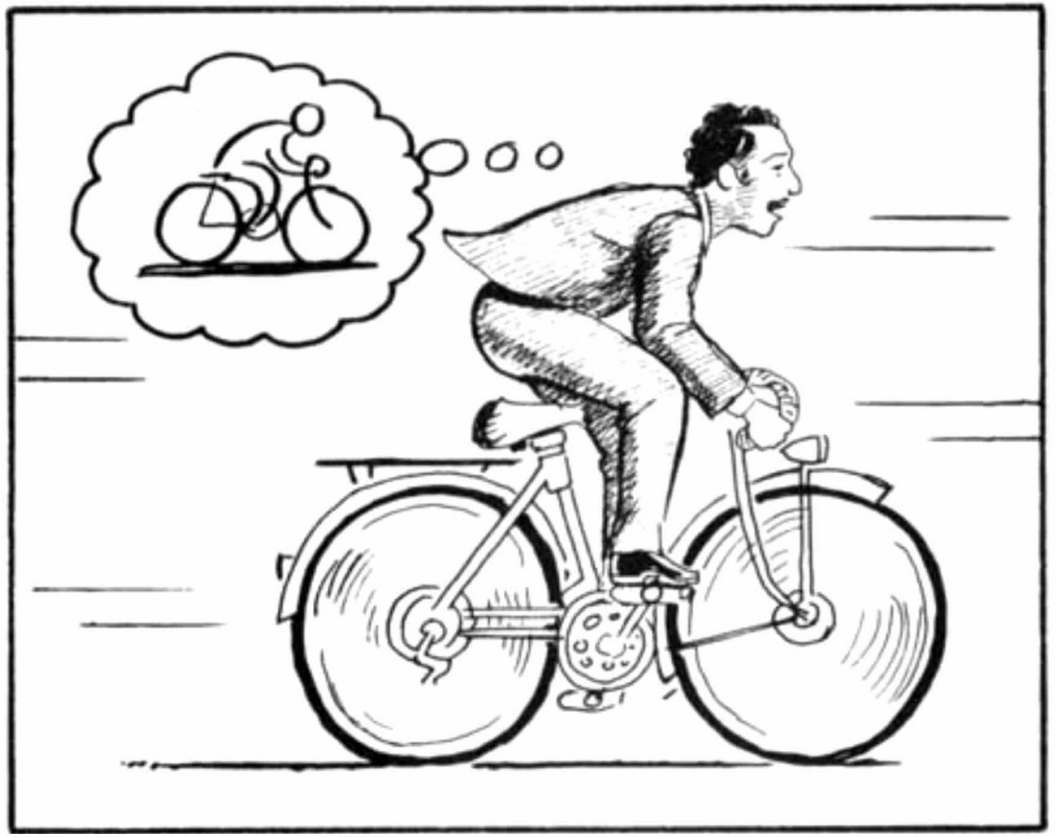




Andy Clark's framework of hierarchical predictive processing models the **brain as an active prediction machine** that minimizes discrepancies between top-down expectations and sensory data.

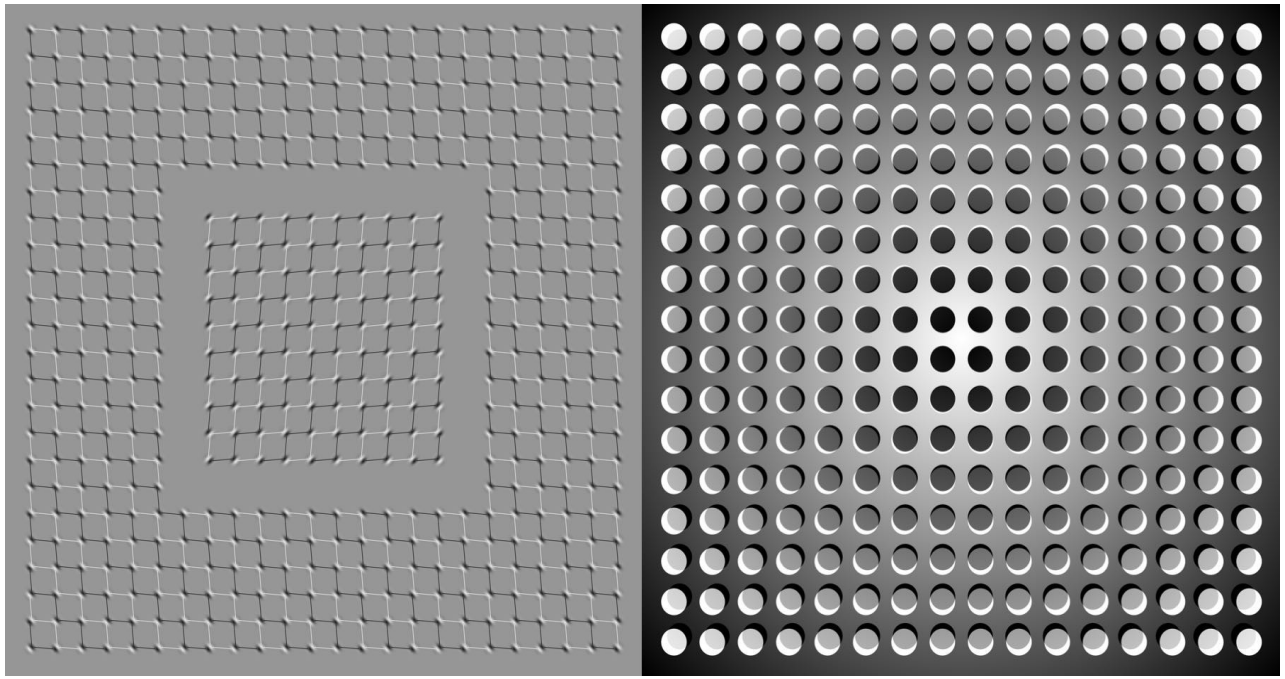
Can agents learn inside of their own dreams?

A model of how the world as we perceive, behaves

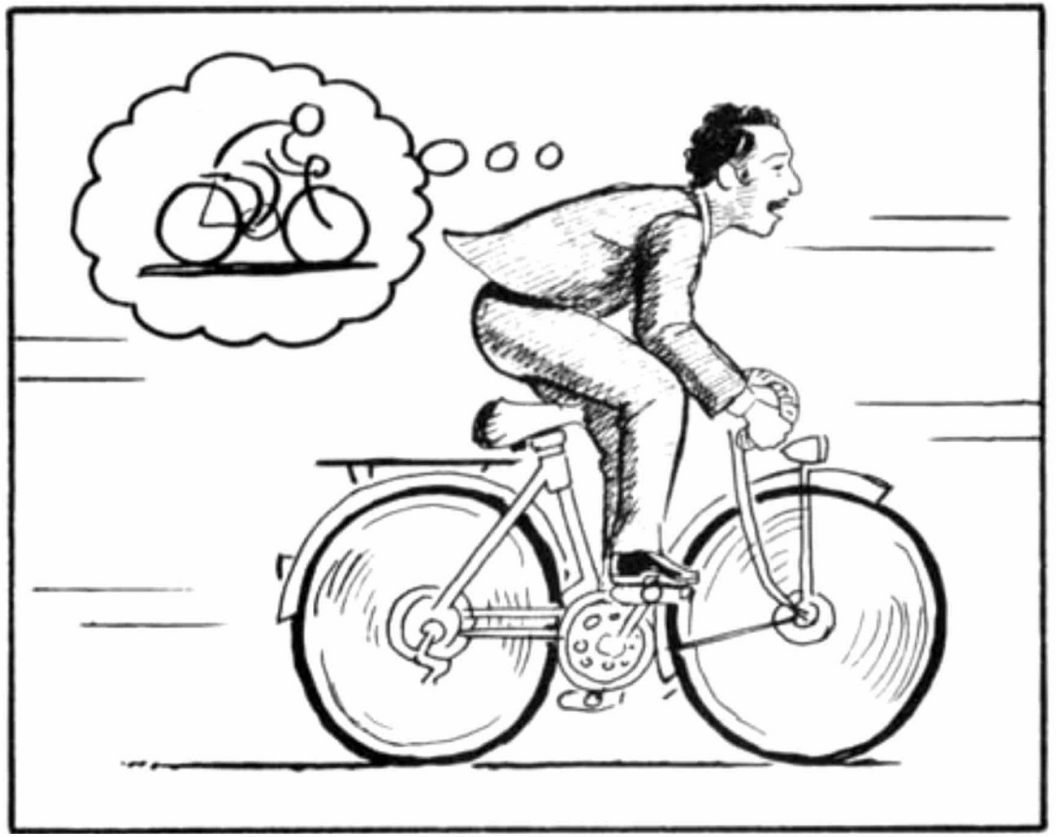


A World Model, from Scott McCloud's *Understanding Comics*.

Can agents learn inside of their own dreams?



What we see is based on our brain's prediction of the future.

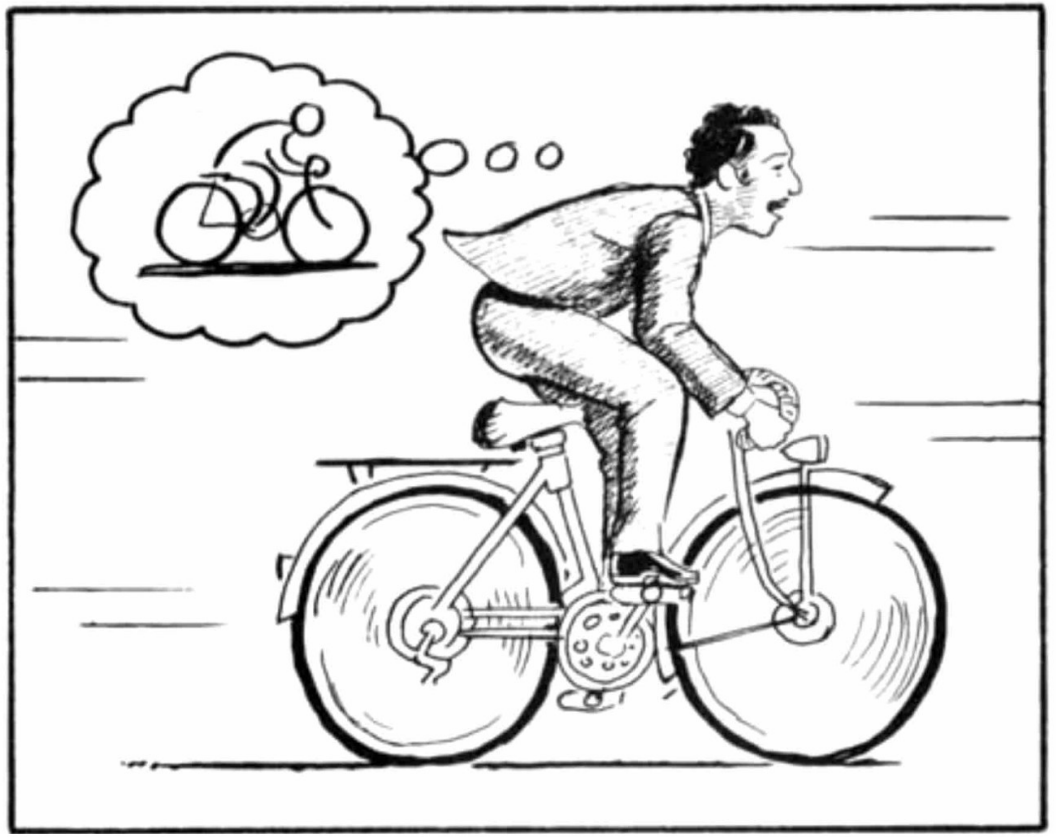


A World Model, from Scott McCloud's *Understanding Comics* 1993

Key Idea

Build a predictive model of the world

Simulate inside this model to understand the effects of actions



A World Model, from Scott McCloud's *Understanding Comics*.

Why Build a World Model?



Predicting the effect of actions is often harder than choosing the *right* actions in the first place

Why Build a World Model? **Generalization+Efficiency**



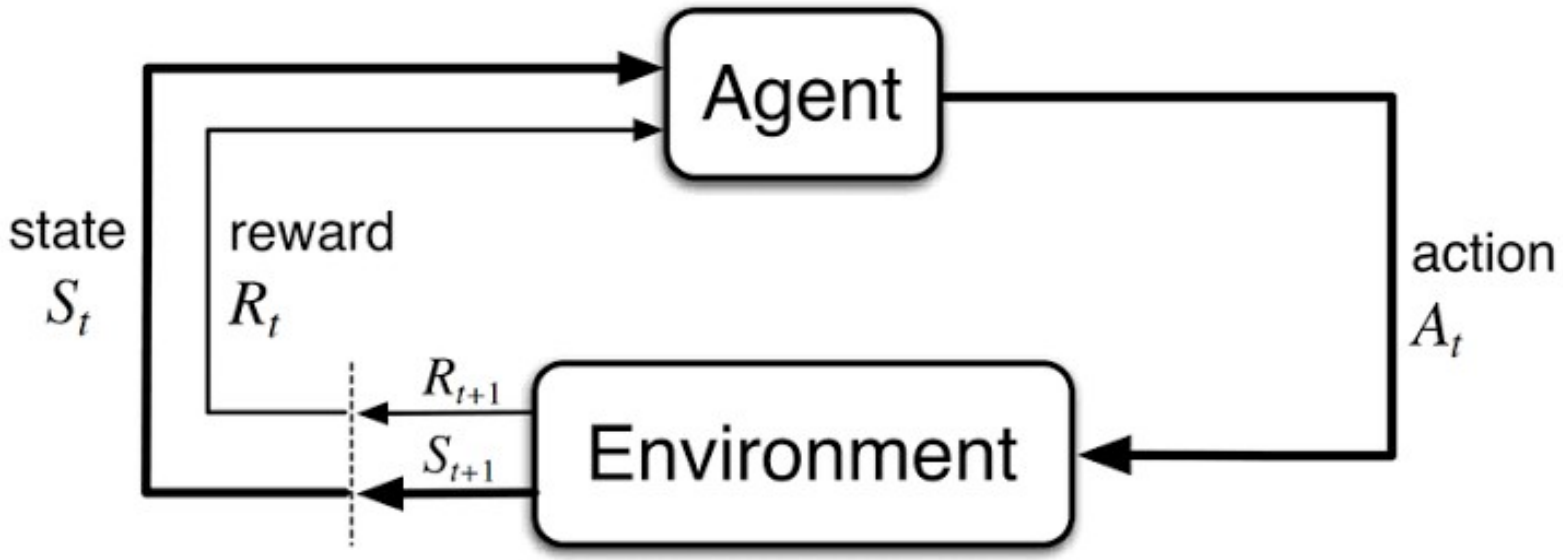
Being able to predict the flow of milk provides useful information about the flow of most other fluids

How to Build a World Model?

Through:

1) Observation

2) Interaction



Towards World Models

Dyna, an Integrated Architecture for Learning, Planning, and Reacting

Richard S. Sutton
GTE Laboratories Incorporated
Waltham, MA 02254
sutton@gte.com



Towards World Models

The Dyna architecture attempts to integrate

- Trial-and-error learning of an optimal *reactive policy*, a mapping from situations to actions;
- Learning of domain knowledge in the form of an *action model*, a black box that takes as input a situation and action and outputs a prediction of the immediate next situation;
- Planning: finding the optimal reactive policy given domain knowledge (the action model);
- Reactive execution: No planning intervenes between perceiving a situation and responding to it.



Towards World Models

The main idea of Dyna is the old, commonsense idea that planning is 'trying things in your head,' using an internal model of the world (Craik, 1943; Dennett, 1978; Sutton & Barto, 1981). This suggests the existence of a more primitive process for trying things *not* in your head, but through direct interaction with the world. *Reinforcement learning* is the name we use for this more primitive, direct kind of trying, and Dyna is the extension of reinforcement learning to include a learned world model.



Towards World Models

REPEAT FOREVER:

1. Observe the world's state and reactively choose an action based on it;
2. Observe resultant reward and new state;
3. Apply reinforcement learning to this experience;
4. Update action model based on this experience;
5. Repeat K times:
   - 5.1 Choose a hypothetical world state and action;
   - 5.2 Predict resultant reward and new state using action model;
   - 5.3 Apply reinforcement learning to this hypothetical experience.

Figure 2: A Generic Dyna Algorithm.



Towards World Models

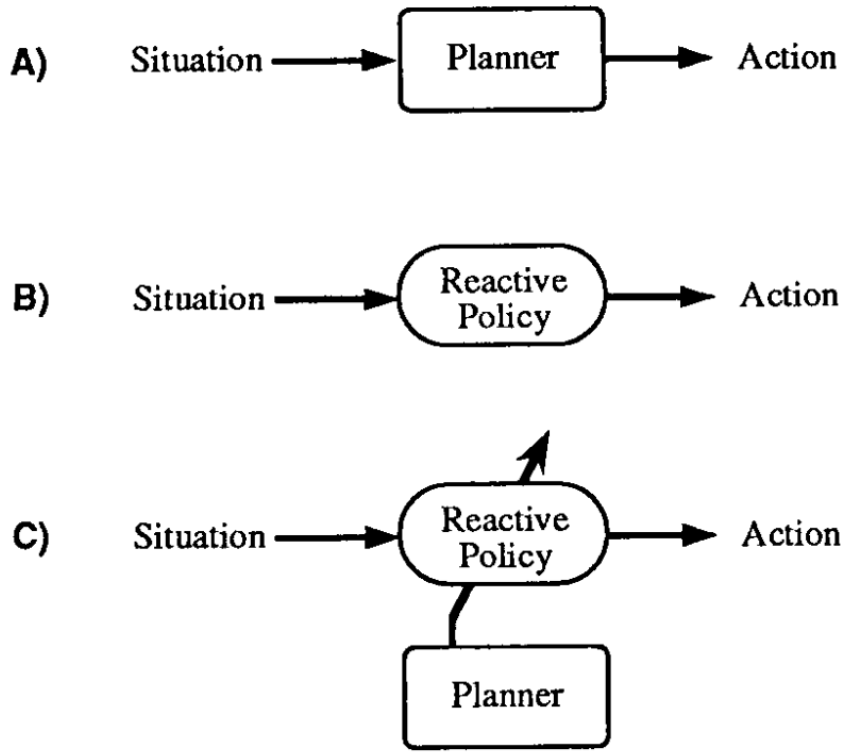
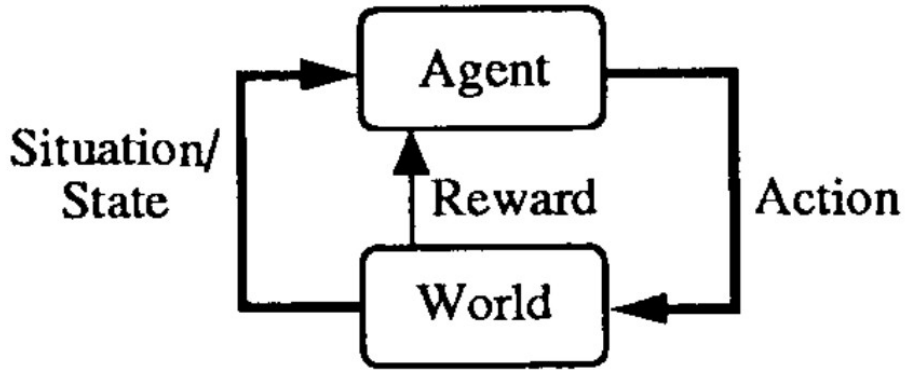


Figure 3: Simplistic Comparison of Architectures: A) Conventional Planning, B) Reactive Systems, C) IDP Planning (incremental compiling into reactions).

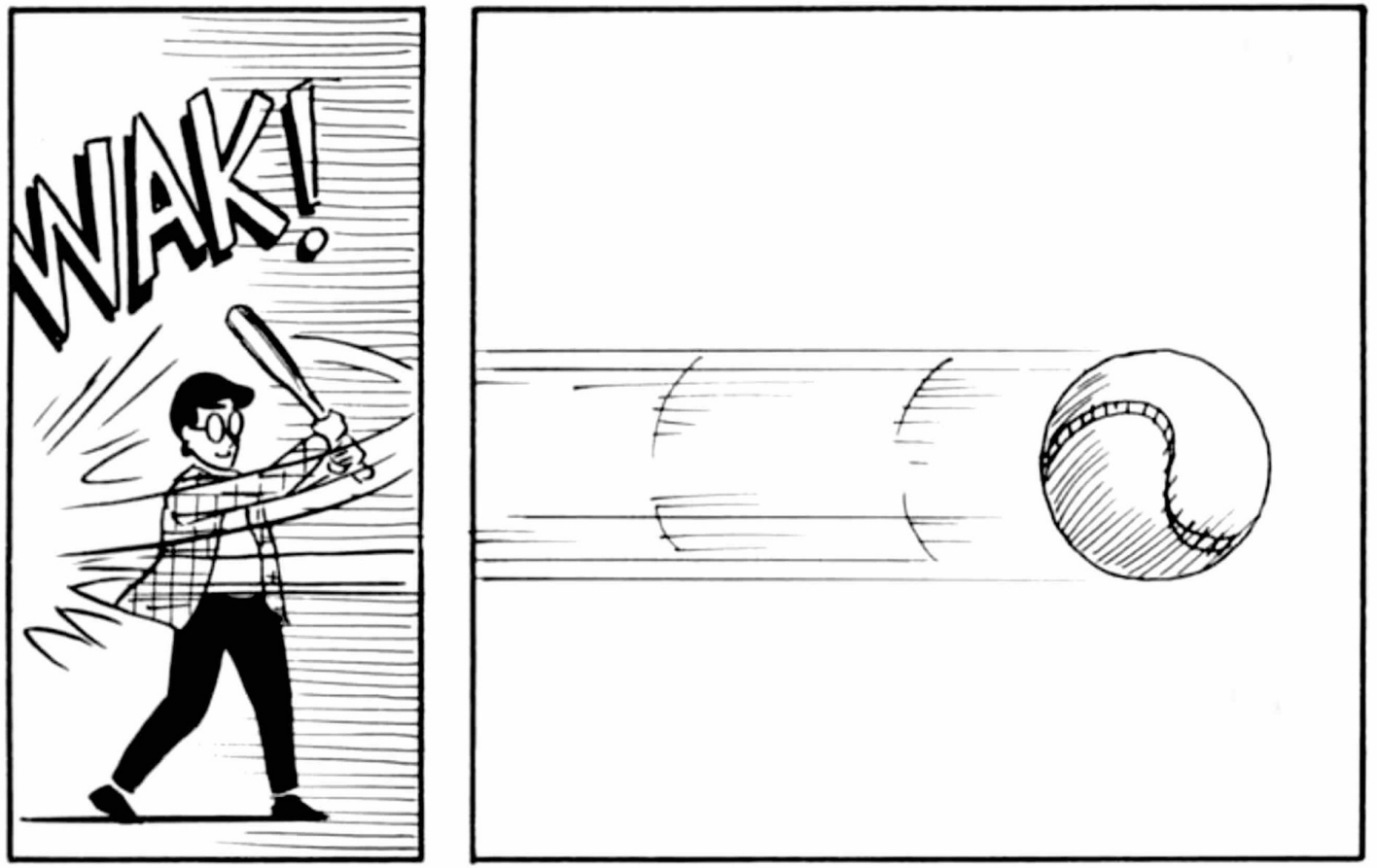


IDP = Incremental Dynamic Programming

Towards World Models

World Models

David Ha [1] **Jürgen Schmidhuber** [2 3]



We learn to perceive time *spatially* when we read comics. According to cartoonist and comics theorist Scott McCloud, "*in the world of comics, time and space are one and the same.*" Art © Scott McCloud. [1]

Towards World Models

At each time step, our agent receives an **observation** from the environment.

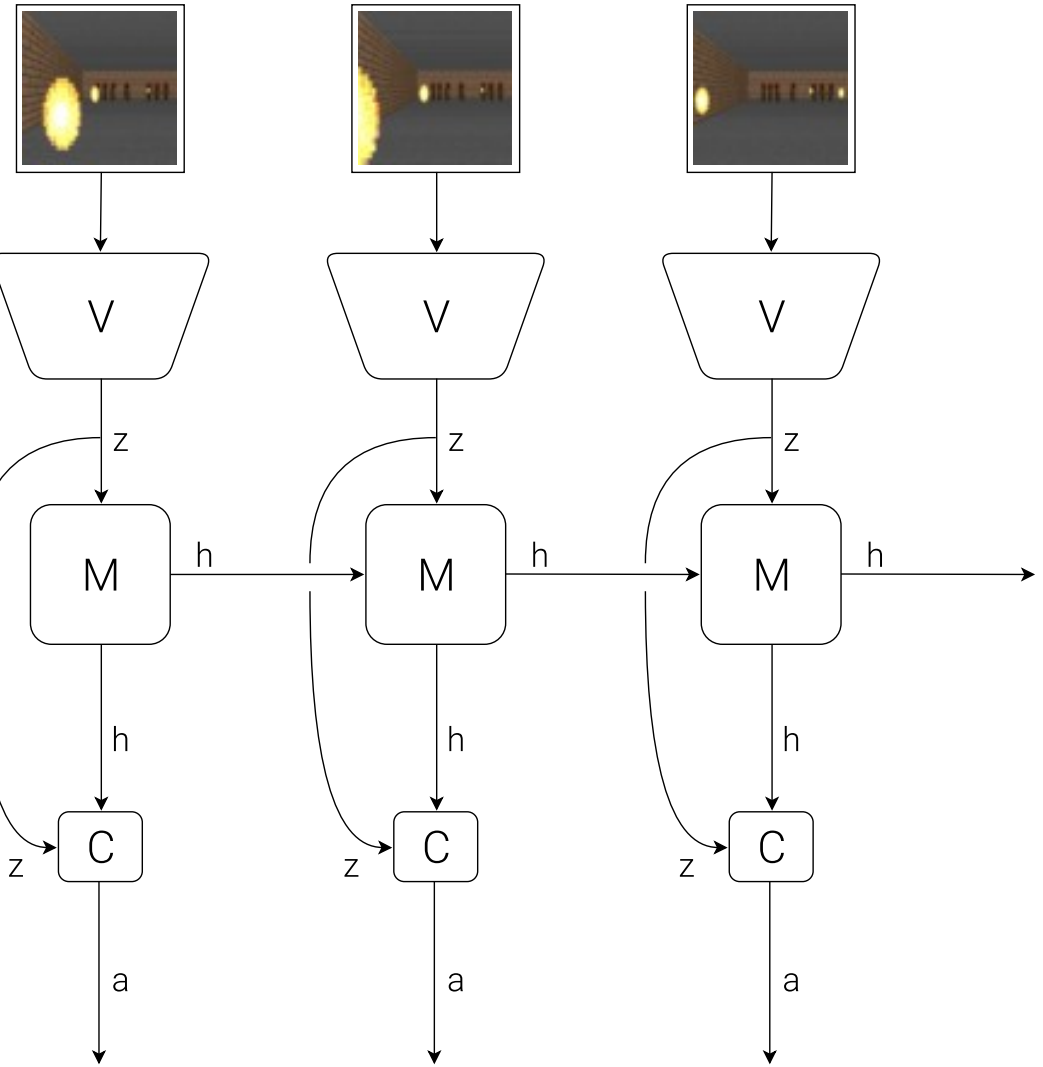
World Model

The **Vision Model (V)** encodes the high-dimensional observation into a low-dimensional latent vector.

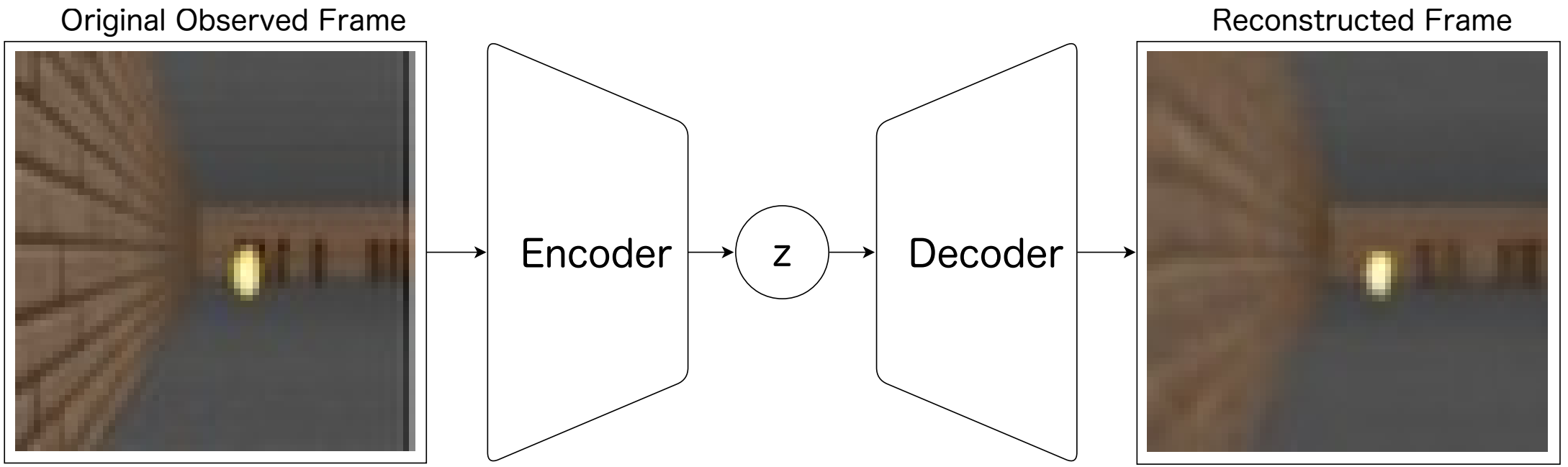
The **Memory RNN (M)** integrates the historical codes to create a representation that can predict future states.

A small **Controller (C)** uses the representations from both **V** and **M** to select good actions.

The agent performs **actions** that go back and affect the environment.

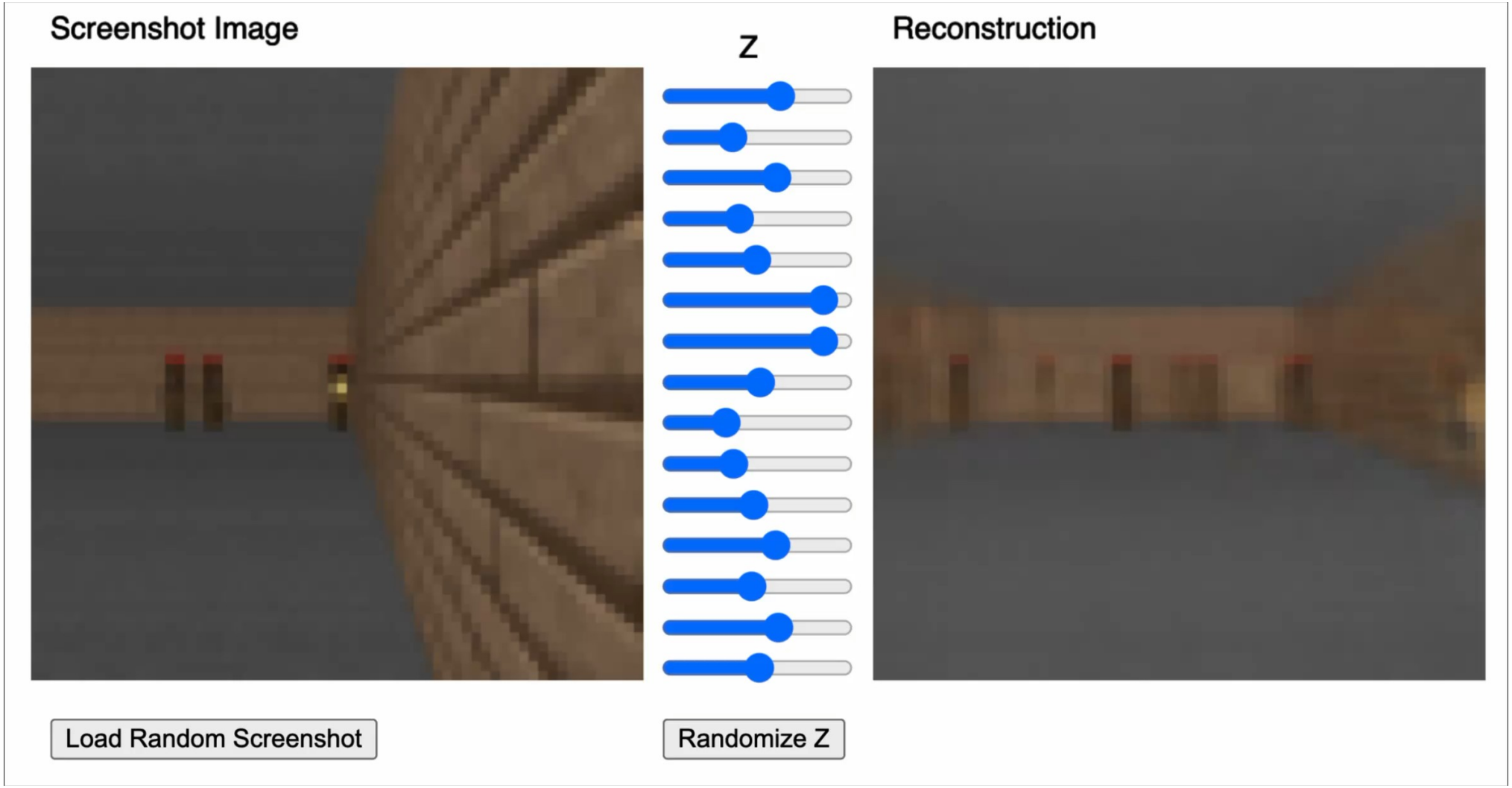


Towards World Models

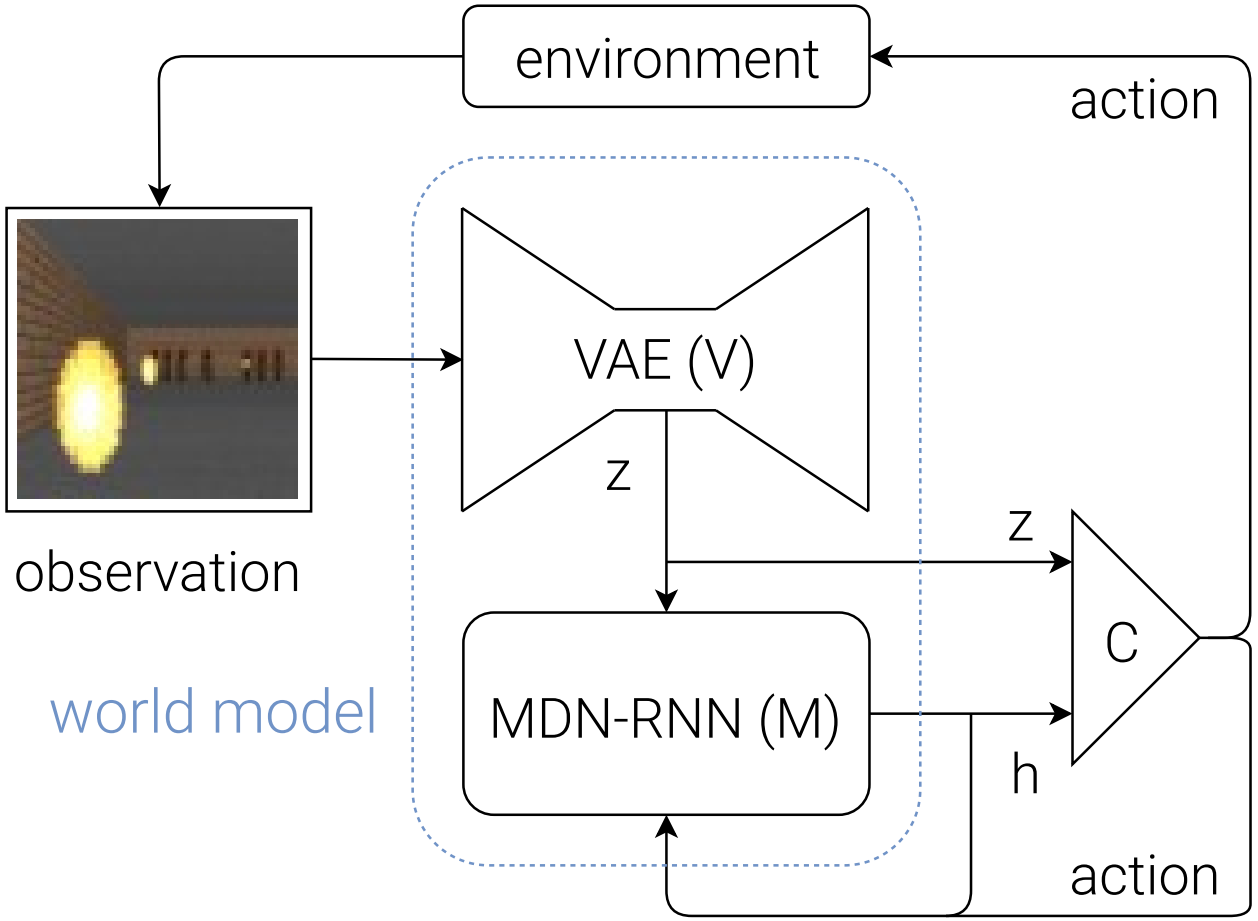


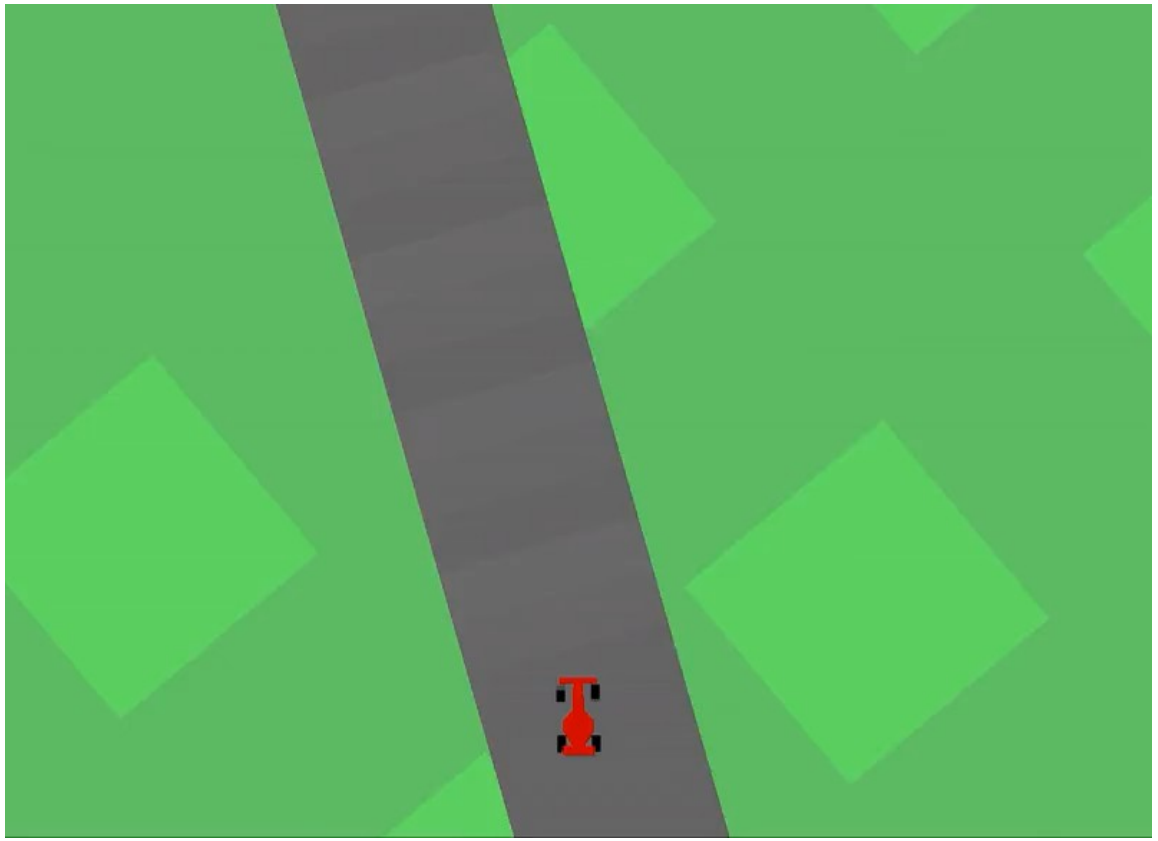
Learning state representations

Towards World Models



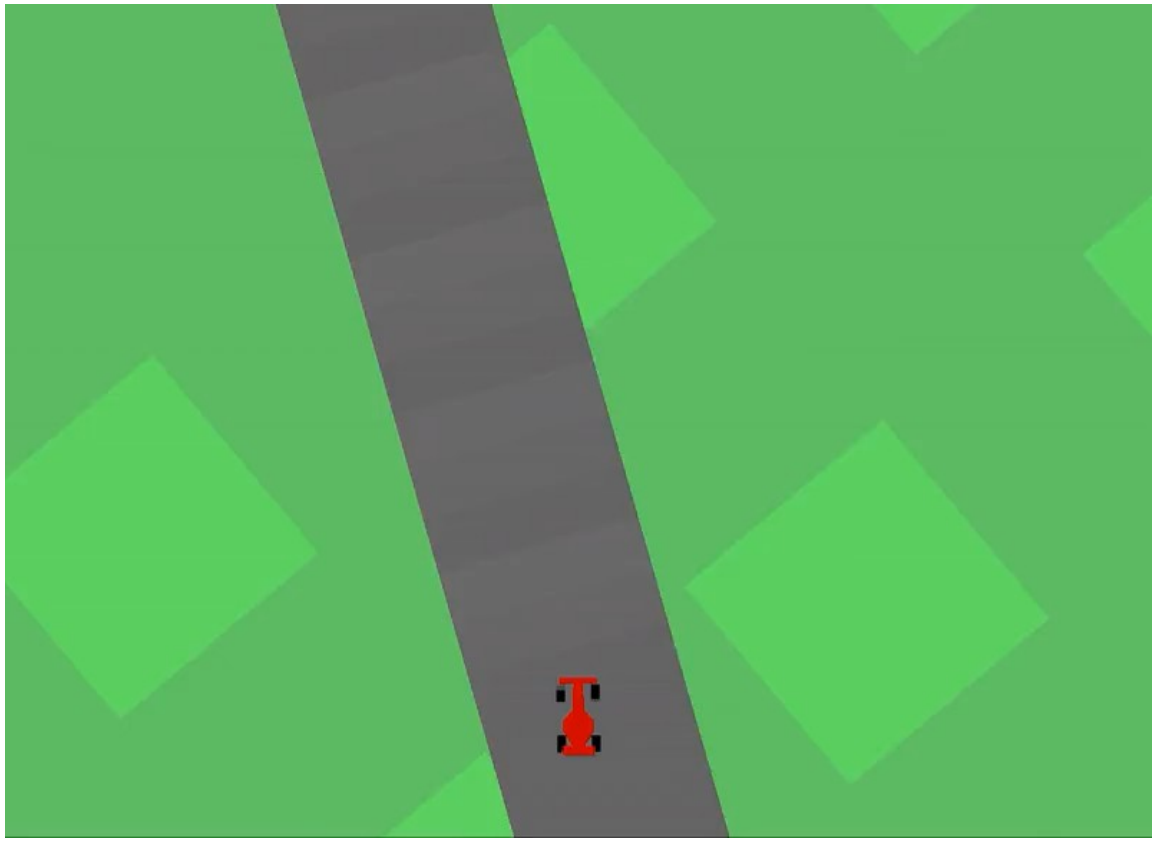
Towards World Models



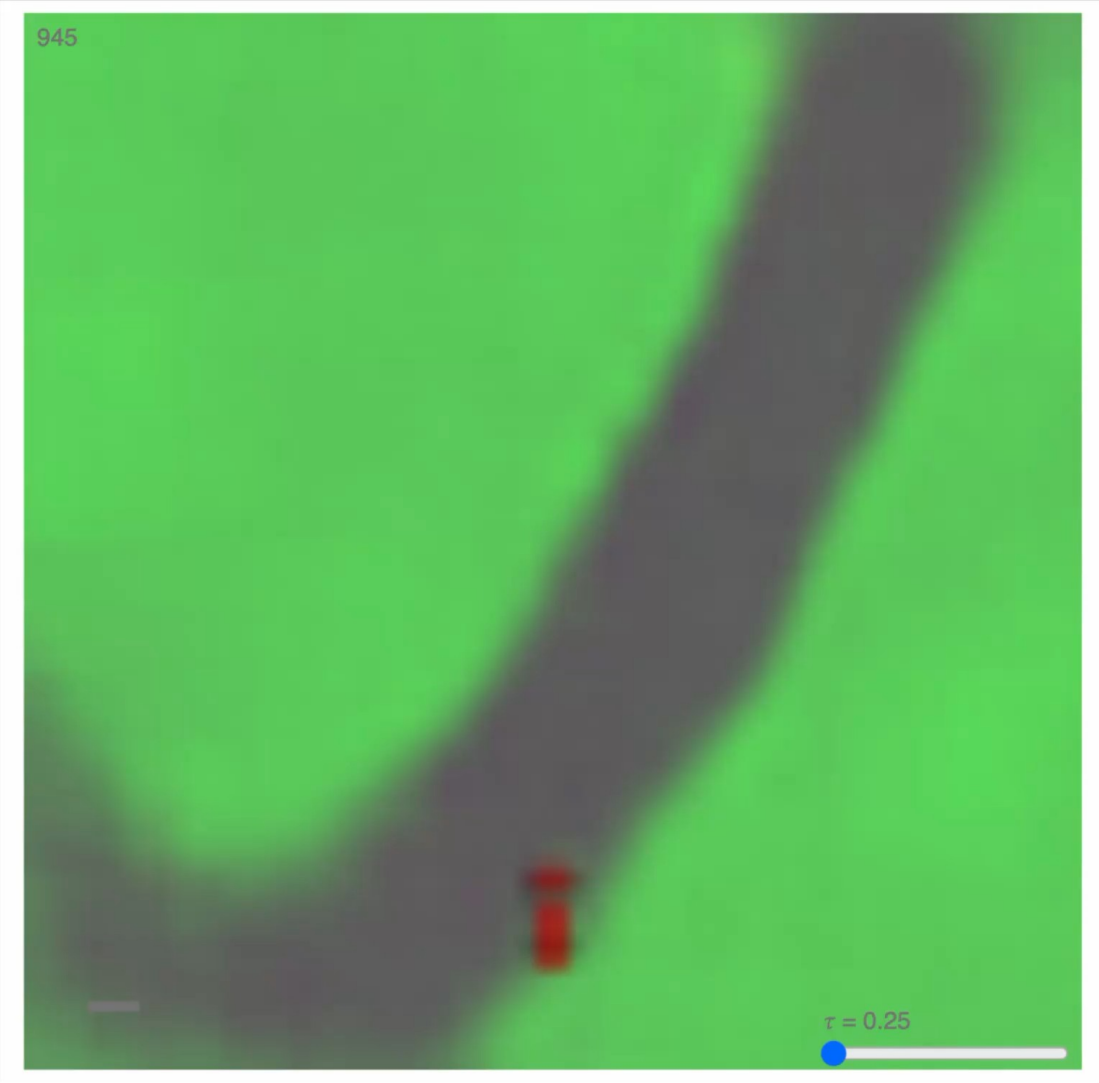


Towards World Models

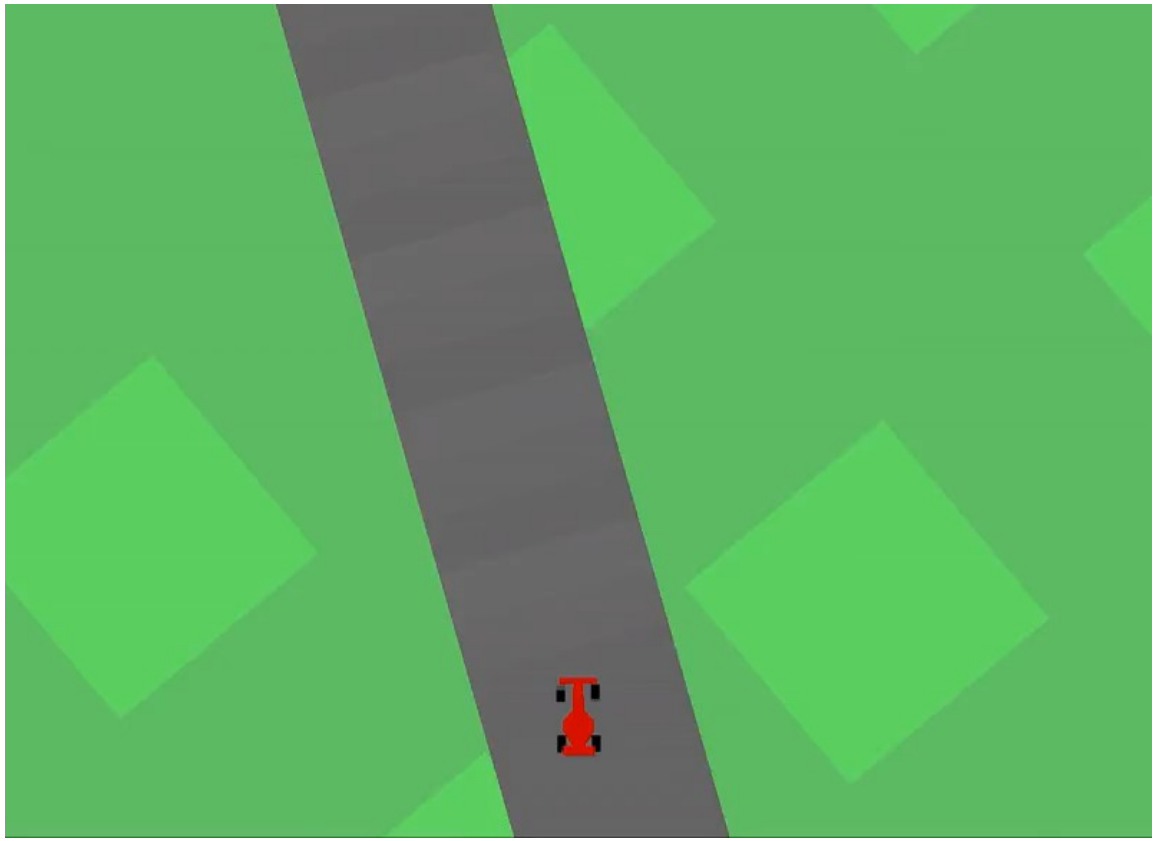
1. Collect 10,000 rollouts from a random policy.
2. Train VAE (V) to encode each frame into a latent vector $z \in \mathcal{R}^{32}$.
3. Train MDN-RNN (M) to model $P(z_{t+1} \mid a_t, z_t, h_t)$.
4. Evolve Controller (C) to maximize the expected cumulative reward of a rollout.



Towards World Models

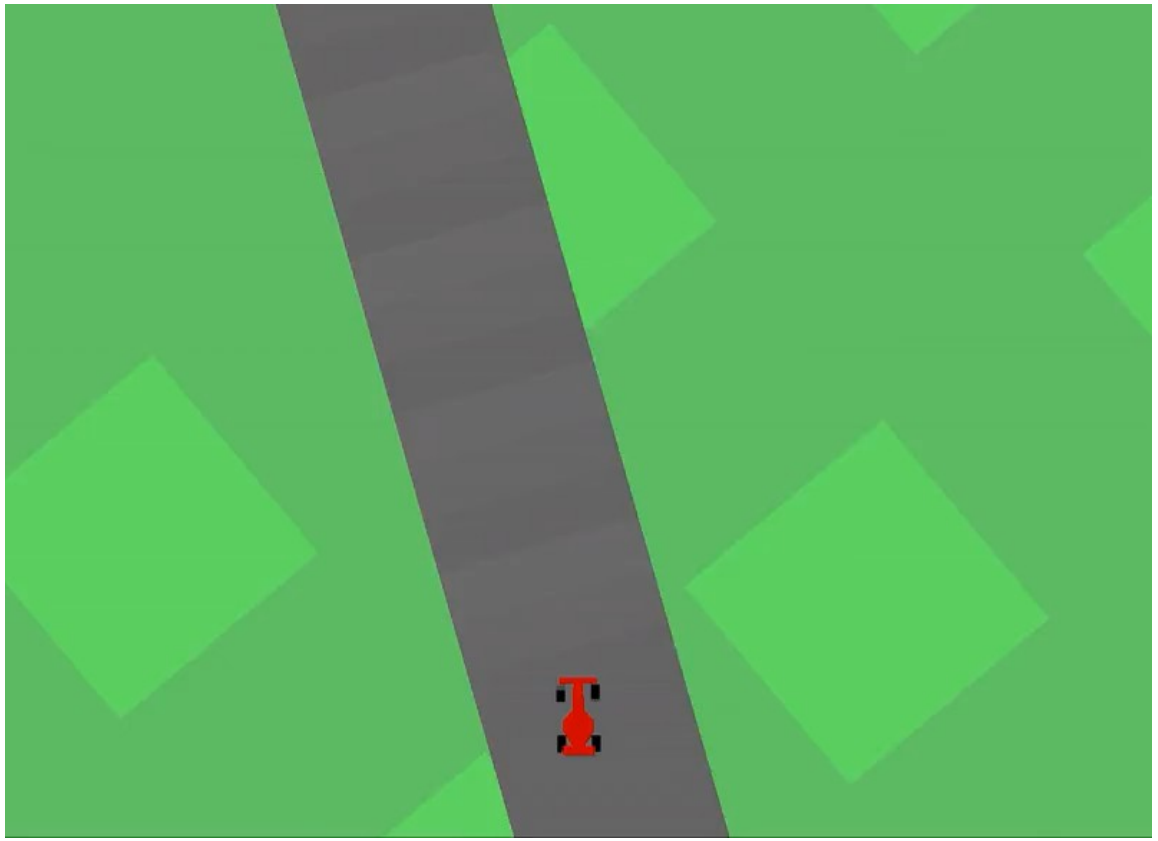


<- Dreaming inside the world model



Towards World Models





Towards World Models

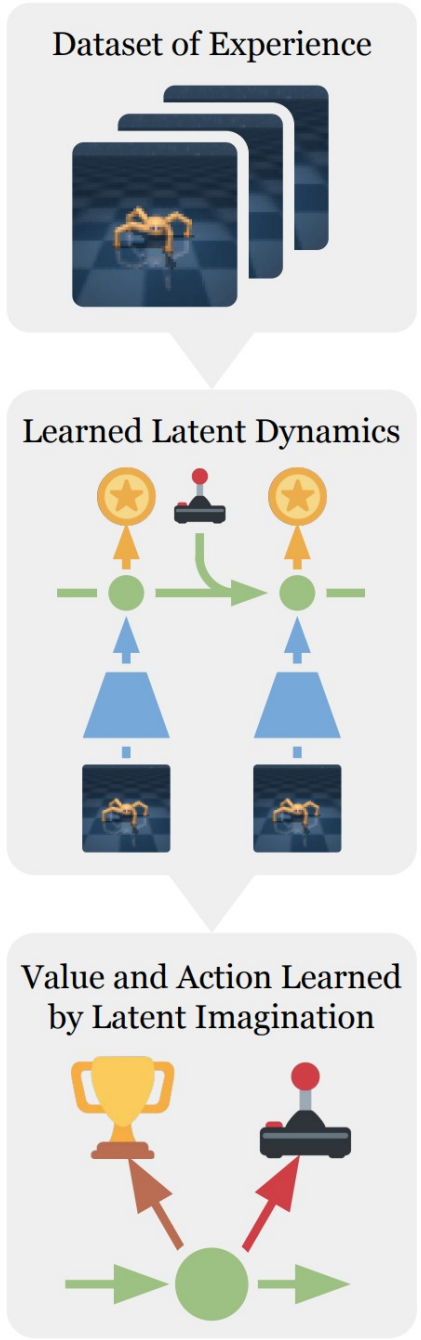
Dream to Control: Learning Behaviors by Latent Imagination

Danijar Hafner *
University of Toronto
Google Brain

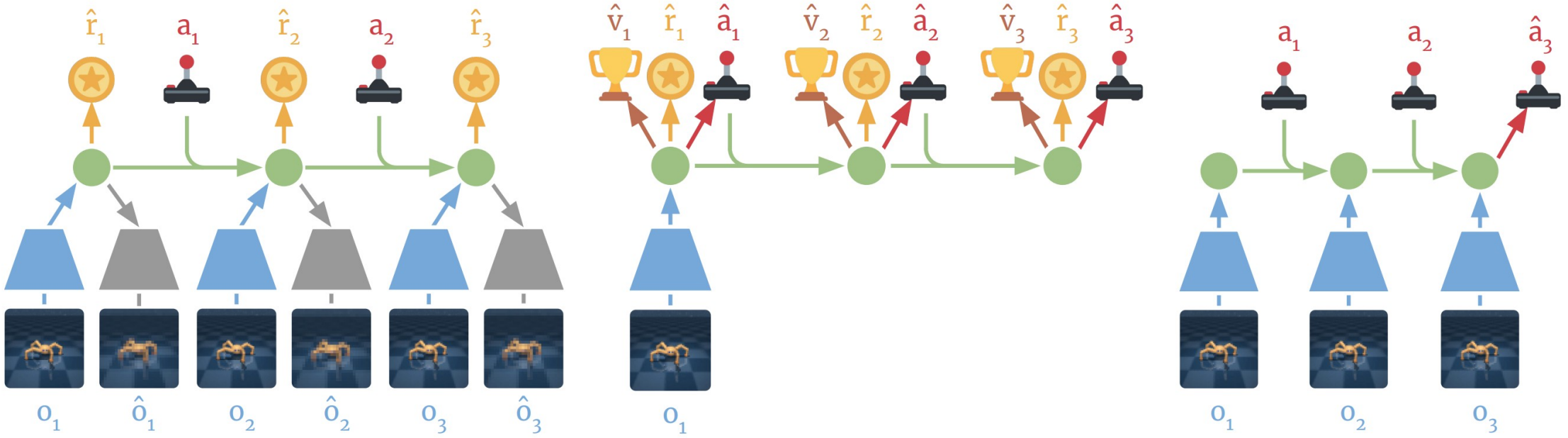
Timothy Lillicrap
DeepMind

Jimmy Ba
University of Toronto

Mohammad Norouzi
Google Brain

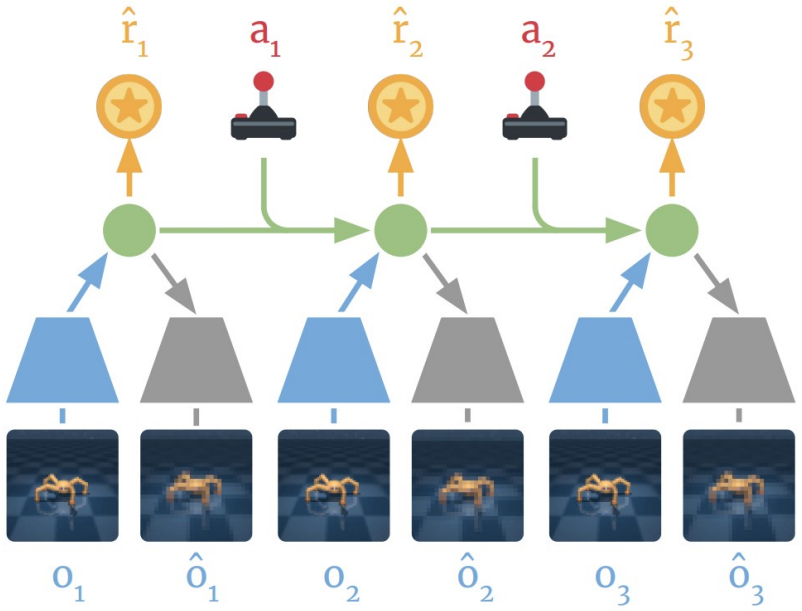


Towards World Models



(a) Learn dynamics from experience (b) Learn behavior in imagination (c) Act in the environment

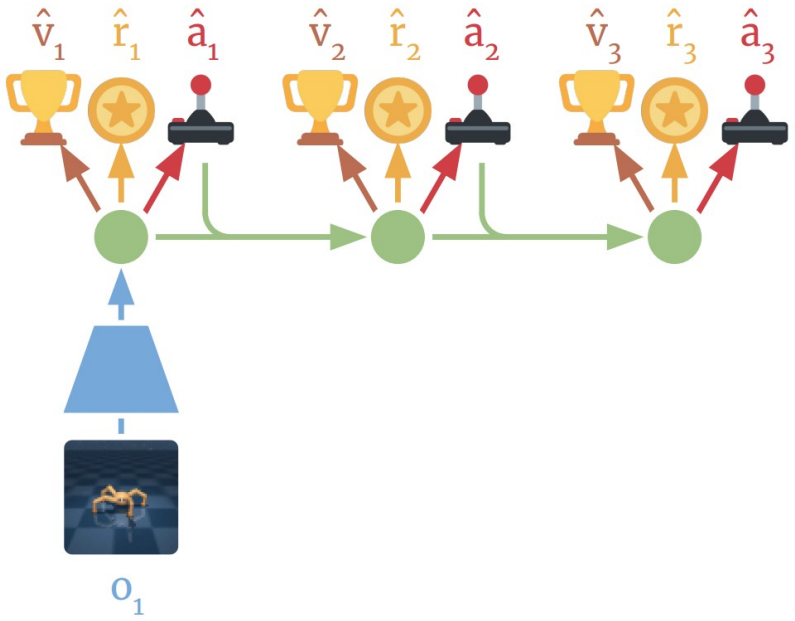
Towards World Models



(a) Learn dynamics from experience

Representation	$p_\theta(s_t \mid s_{t-1}, a_{t-1}, o_t)$
Transition	$q_\theta(s_t \mid s_{t-1}, a_{t-1})$
Reward	$q_\theta(r_t \mid s_t)$

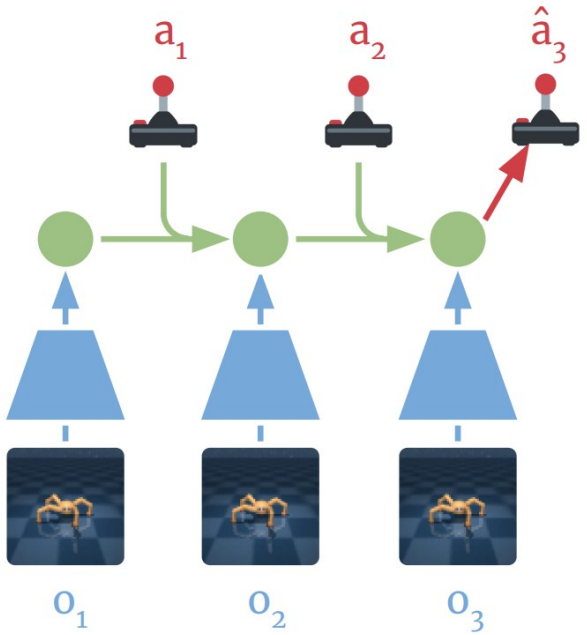
Towards World Models



(b) Learn behavior in imagination

Representation	$p_\theta(s_t \mid s_{t-1}, a_{t-1}, o_t)$
Transition	$q_\theta(s_t \mid s_{t-1}, a_{t-1})$
Reward	$q_\theta(r_t \mid s_t)$
Action	$q_\phi(a_t \mid s_t)$
Value	$v_\psi(s_t)$

Towards World Models



(c) Act in the environment

Representation	$p_\theta(s_t \mid s_{t-1}, a_{t-1}, o_t)$
Transition	$q_\theta(s_t \mid s_{t-1}, a_{t-1})$
Reward	$q_\theta(r_t \mid s_t)$
Action	$q_\phi(a_t \mid s_t)$
Value	$v_\psi(s_t)$

$$a_\tau \sim q_\phi(a_\tau \mid s_\tau)$$

$$v_\psi(s_\tau) \approx \mathrm{E}_{q(\cdot \mid s_\tau)}\left(\sum_{\tau=t}^{t+H} \gamma^{\tau-t} r_\tau\right)$$

Towards World Models

Algorithm 1: Dreamer

Initialize dataset $\mathcal{D}$ with S random seed episodes.
Initialize neural network parameters θ, ϕ, ψ randomly.
while *not converged* **do**
  for *update step* $c = 1..C$ **do**
    `// Dynamics learning`
    Draw B data sequences $\{(a_t, o_t, r_t)\}_{t=k}^{k+L} \sim \mathcal{D}$.
    Compute model states $s_t \sim p_\theta(s_t \mid s_{t-1}, a_{t-1}, o_t)$.
    Update θ using representation learning.
    `// Behavior learning`
    Imagine trajectories $\{(s_\tau, a_\tau)\}_{\tau=t}^{t+H}$ from each s_t.
    Predict rewards $\mathrm{E}\big(q_\theta(r_\tau \mid s_\tau)\big)$ and values $v_\psi(s_\tau)$.
    Compute value estimates $\mathrm{V}_\lambda(s_\tau)$ via Equation 6.
    Update $\phi \leftarrow \phi + \alpha \nabla_\phi \sum_{\tau=t}^{t+H} \mathrm{V}_\lambda(s_\tau)$.
    Update $\psi \leftarrow \psi - \alpha \nabla_\psi \sum_{\tau=t}^{t+H} \frac{1}{2} \big\| v_\psi(s_\tau) - \mathrm{V}_\lambda(s_\tau) \big\|^2$.
  `// Environment interaction`
  $o_1 \leftarrow$ `env.reset()`
  for *time step* $t = 1..T$ **do**
    Compute $s_t \sim p_\theta(s_t \mid s_{t-1}, a_{t-1}, o_t)$ from history.
    Compute $a_t \sim q_\phi(a_t \mid s_t)$ with the action model.
    Add exploration noise to action.
    $r_t, o_{t+1} \leftarrow$ `env.step(`a_t`)`.
  Add experience to dataset $\mathcal{D} \leftarrow \mathcal{D} \cup \{(o_t, a_t, r_t)_{t=1}^{T}\}$.

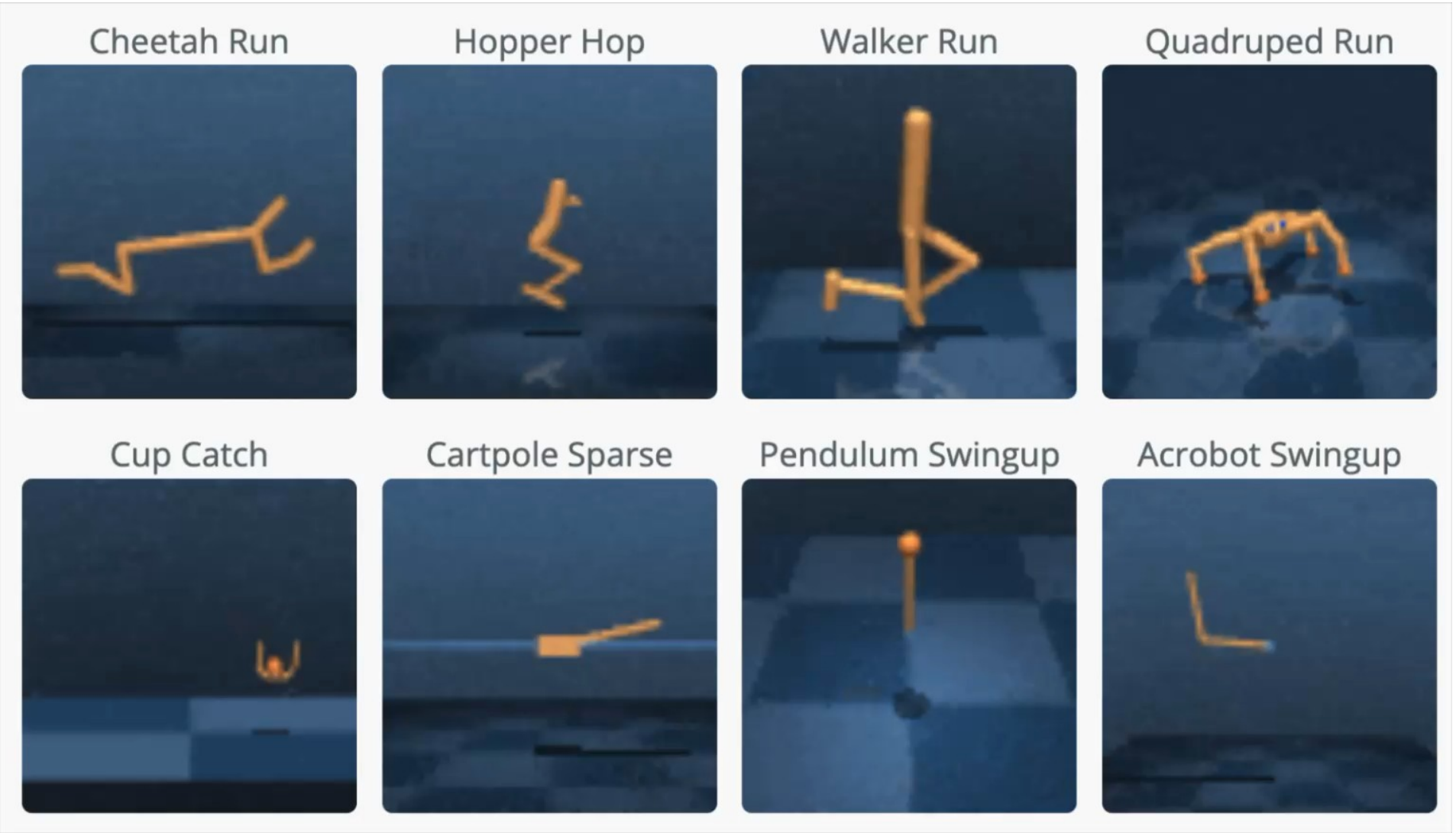
Model components

Representation	$p_\theta(s_t \mid s_{t-1}, a_{t-1}, o_t)$
Transition	$q_\theta(s_t \mid s_{t-1}, a_{t-1})$
Reward	$q_\theta(r_t \mid s_t)$
Action	$q_\phi(a_t \mid s_t)$
Value	$v_\psi(s_t)$

Hyper parameters

Seed episodes	S
Collect interval	C
Batch size	B
Sequence length	L
Imagination horizon	H
Learning rate	α

Towards World Models



Towards World Models

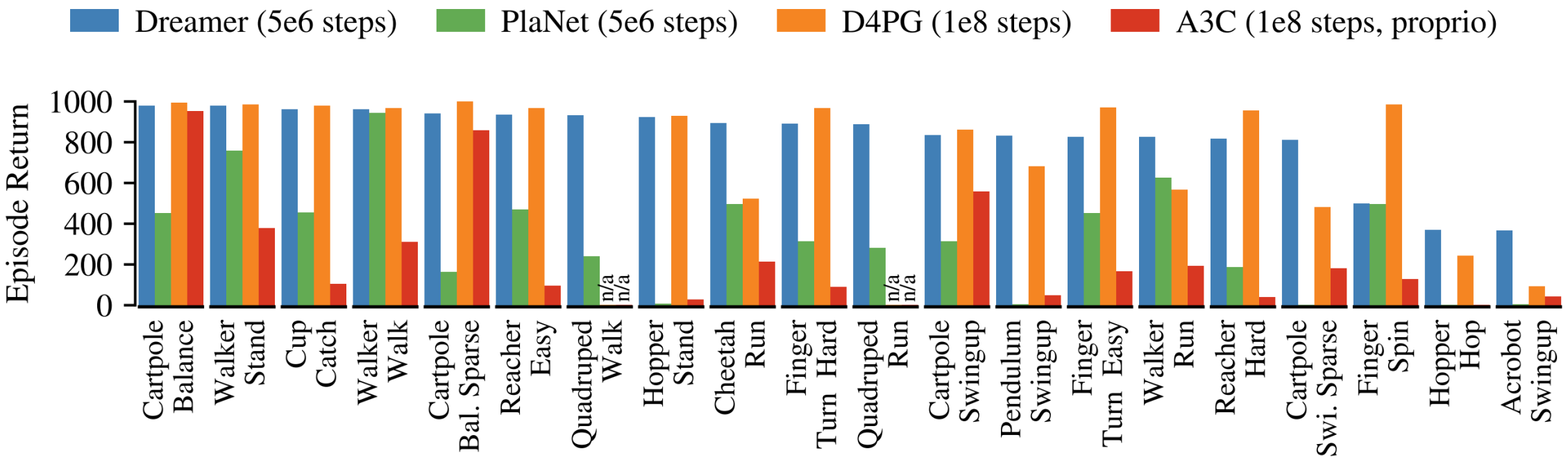
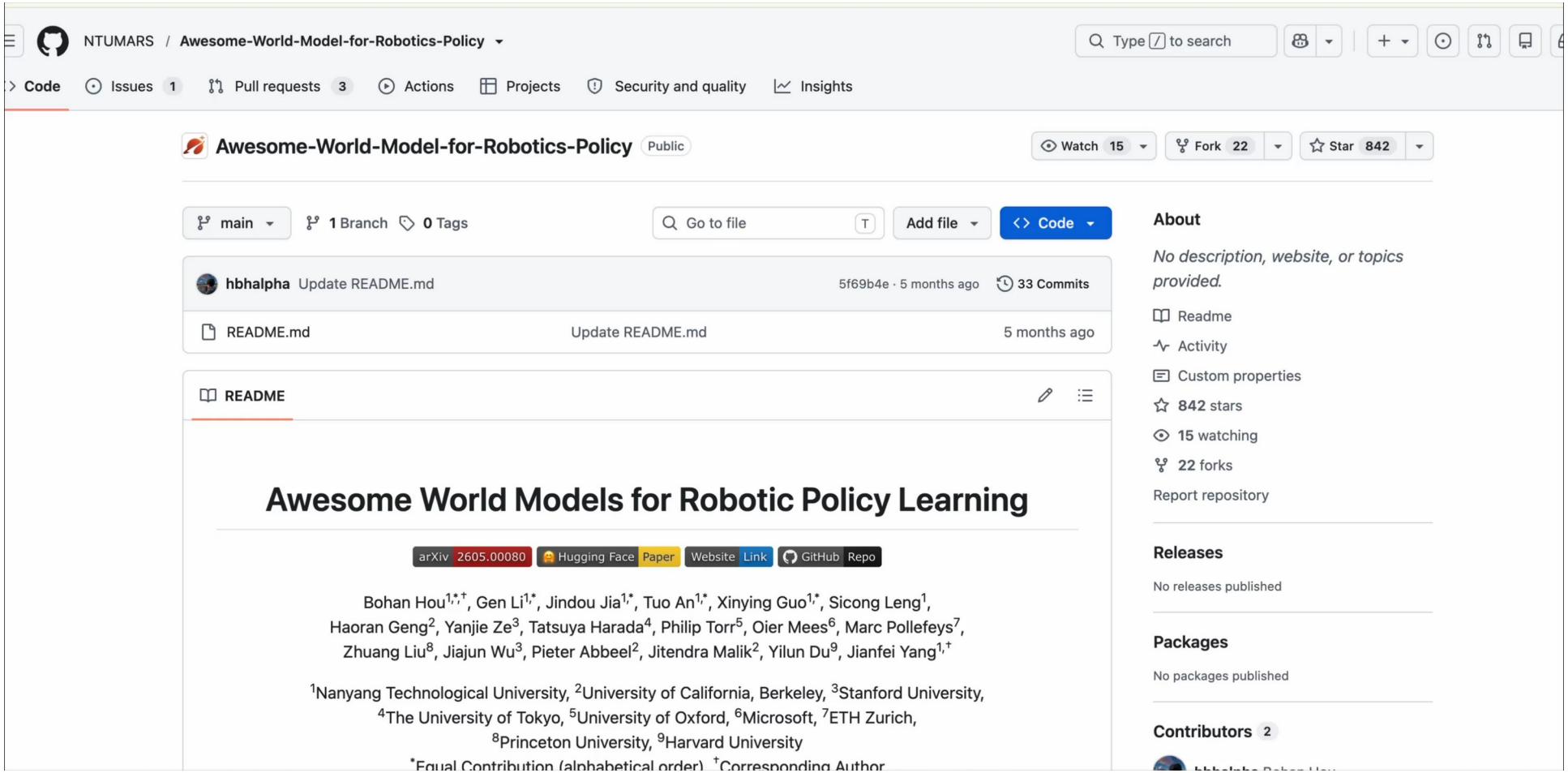


Figure 6: Performance comparison to existing methods. Dreamer inherits the data-efficiency of PlaNet while exceeding the asymptotic performance of the best model-free agents. After 5×10^6 environment steps, Dreamer reaches an average performance of 823 across tasks, compared to PlaNet at 332 and the top model-free D4PG agent at 786 after 10^8 steps. Results are averages over 5 seeds.

World Models from *Offline* Data



If you squint hard enough, all models are world models ☺

World Models from *Offline* Data

Claim: Just image conditioned video generation is NOT world modeling

WHY?

World Models from *Offline* Data

Claim: Just image conditioned video generation is NOT world modeling

WHY?

There are no “actions” to observe the effects of

World Models from *Offline* Data

Claim: Just image conditioned video generation is NOT world modeling

WHY?

There are no “actions” to observe the effects of

How about image+text conditioned video generation?

World Models from *Offline* Data

Claim: Just image conditioned video generation is NOT world modeling

WHY?

There are no “actions” to observe the effects of

How about image+text conditioned video generation? Depends on how much you want to squint

World Models from *Offline* Data

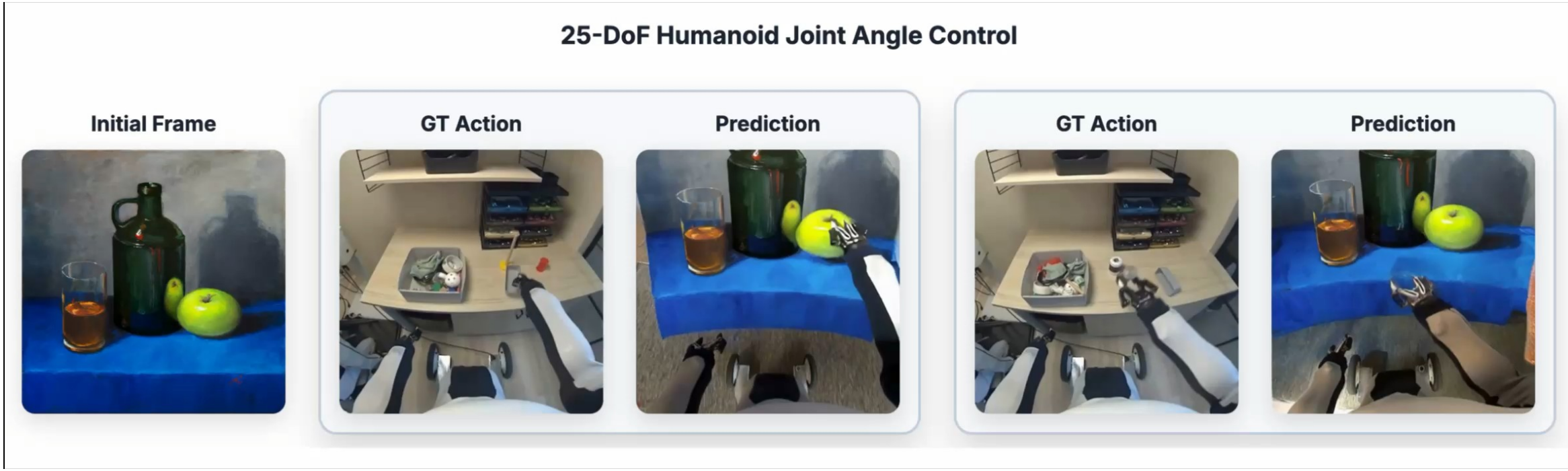
Walk Through Paintings : Ego-centric World Models from Internet Priors

Anurag Bagchi[1], Zhipeng Bao[1], Homanga Bharadhwaj[1], Yu-Xiong Wang[2], Pavel Tokmakov[3], Martial Hebert[1]
[1]Carnegie Mellon University, [2]UIUC, [3]TRI

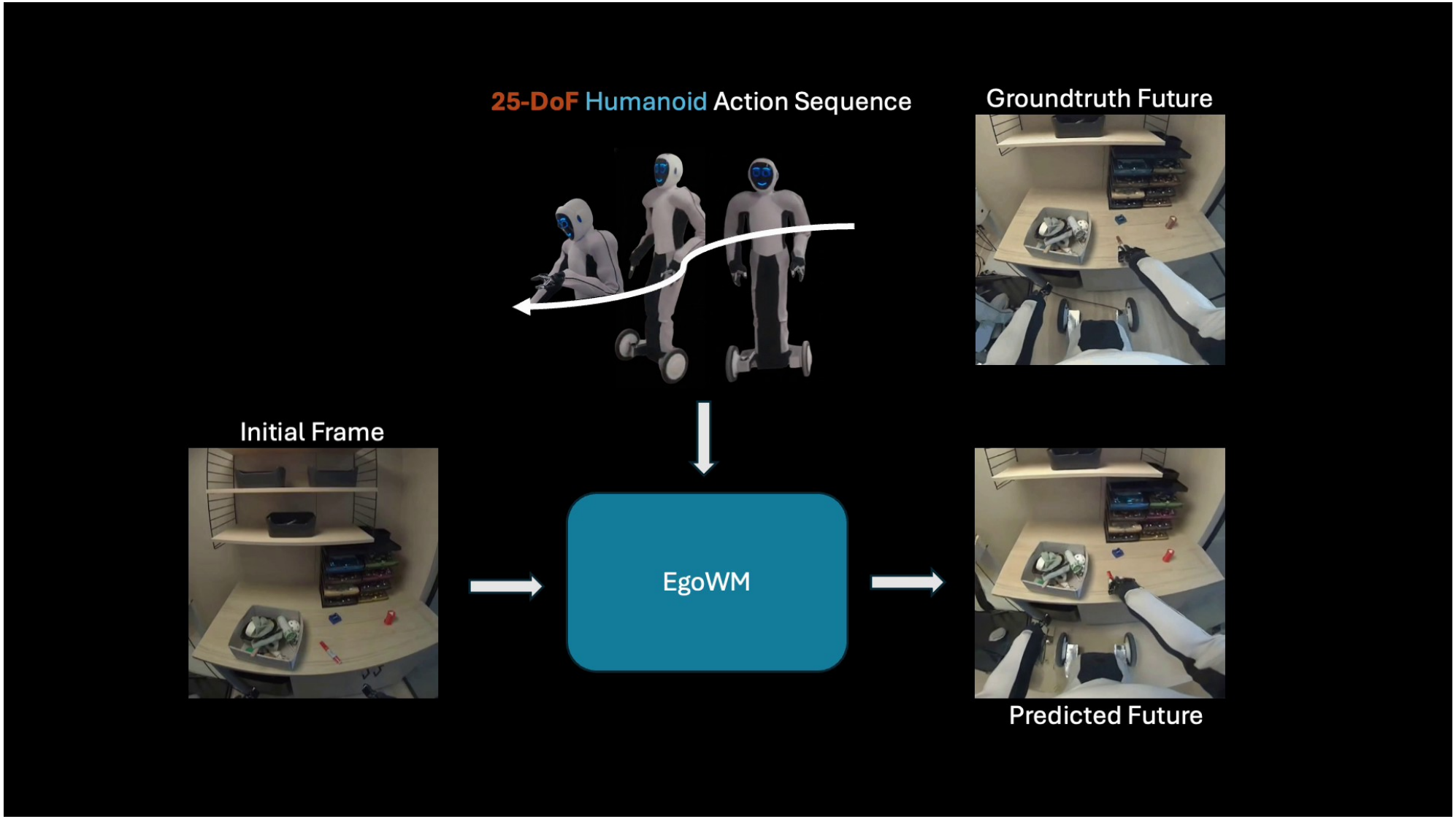
ECCV 2026

arXiv Code

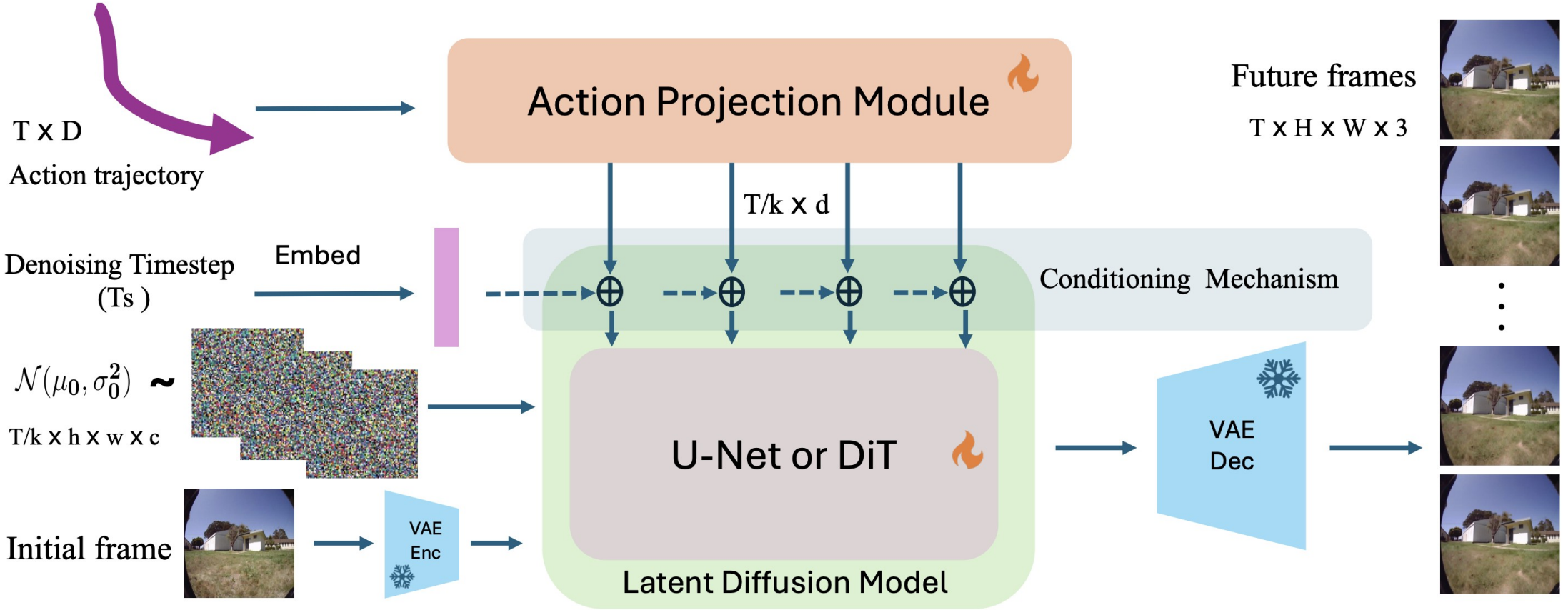
TL;DR Initial Frame (I_0) + Action Trajectory ($A_{1:T}$) → Future Frames ($I_{1:T}$)



World Models from *Offline* Data



World Models from *Offline* Data



World Models from *Offline* Data

What 30,000 hours of ego-centric video does not teach

Jiahua Dong [1] Anurag Bagchi [2] Yash Jangir [3] Muhammad Zubair Irshad [4] Sergey Zakharov [4] Martial Hebert [2] Homanga Bharadhwaj [3] Yu-Xiong Wang [1]
Vitor Campagnolo Guizilini [4] Pavel Tokmakov [4]

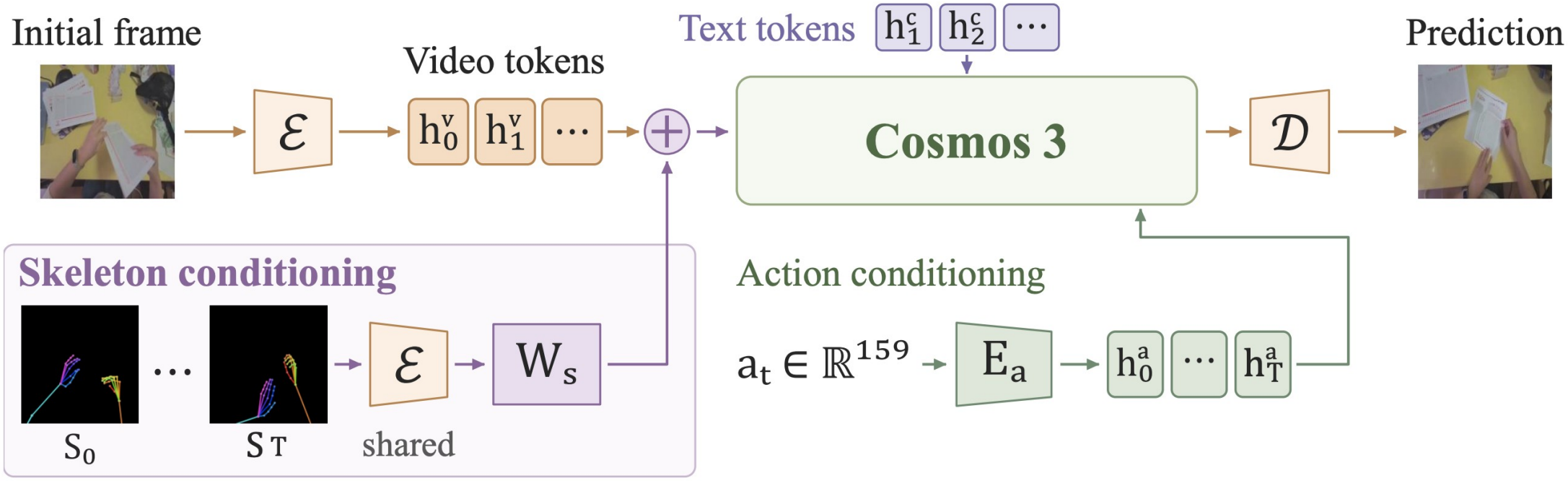
[1] ILLINOIS [2] Carnegie Mellon University [3] JOHNS HOPKINS UNIVERSITY [4] TOYOTA RESEARCH INSTITUTE

DATA PARTNER mecka

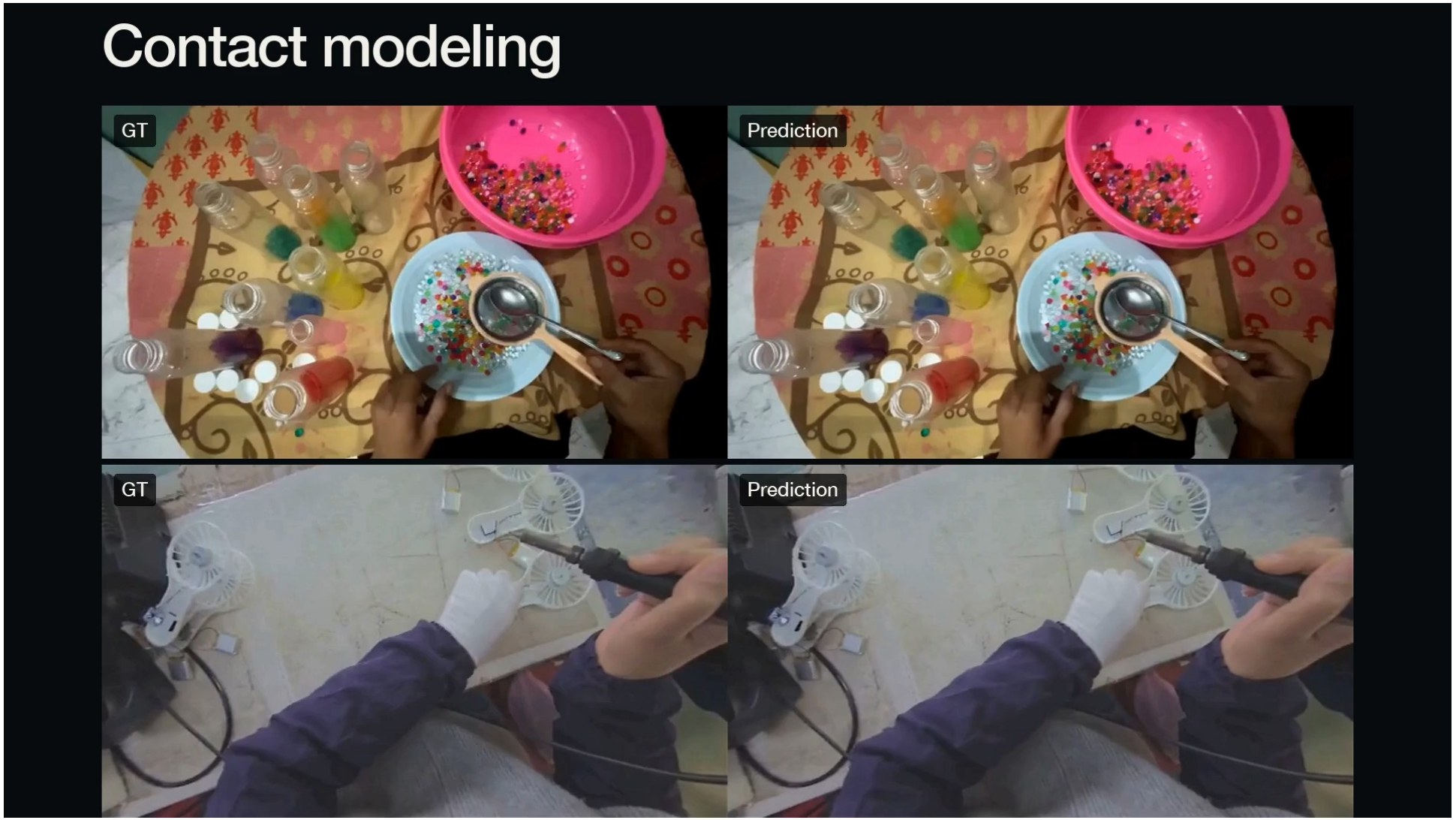
World Models from *Offline* Data



World Models from *Offline* Data

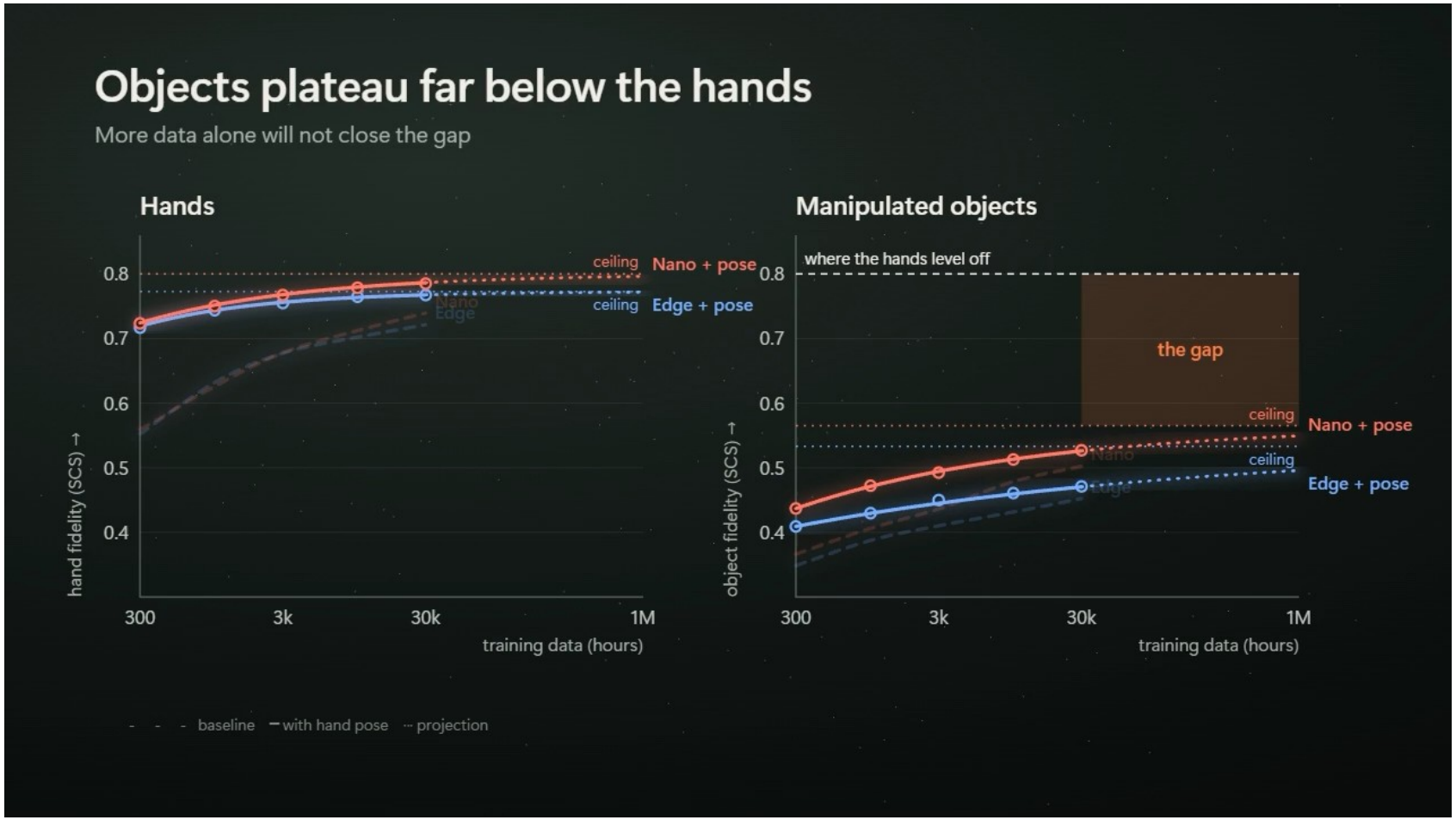


World Models from *Offline* Data



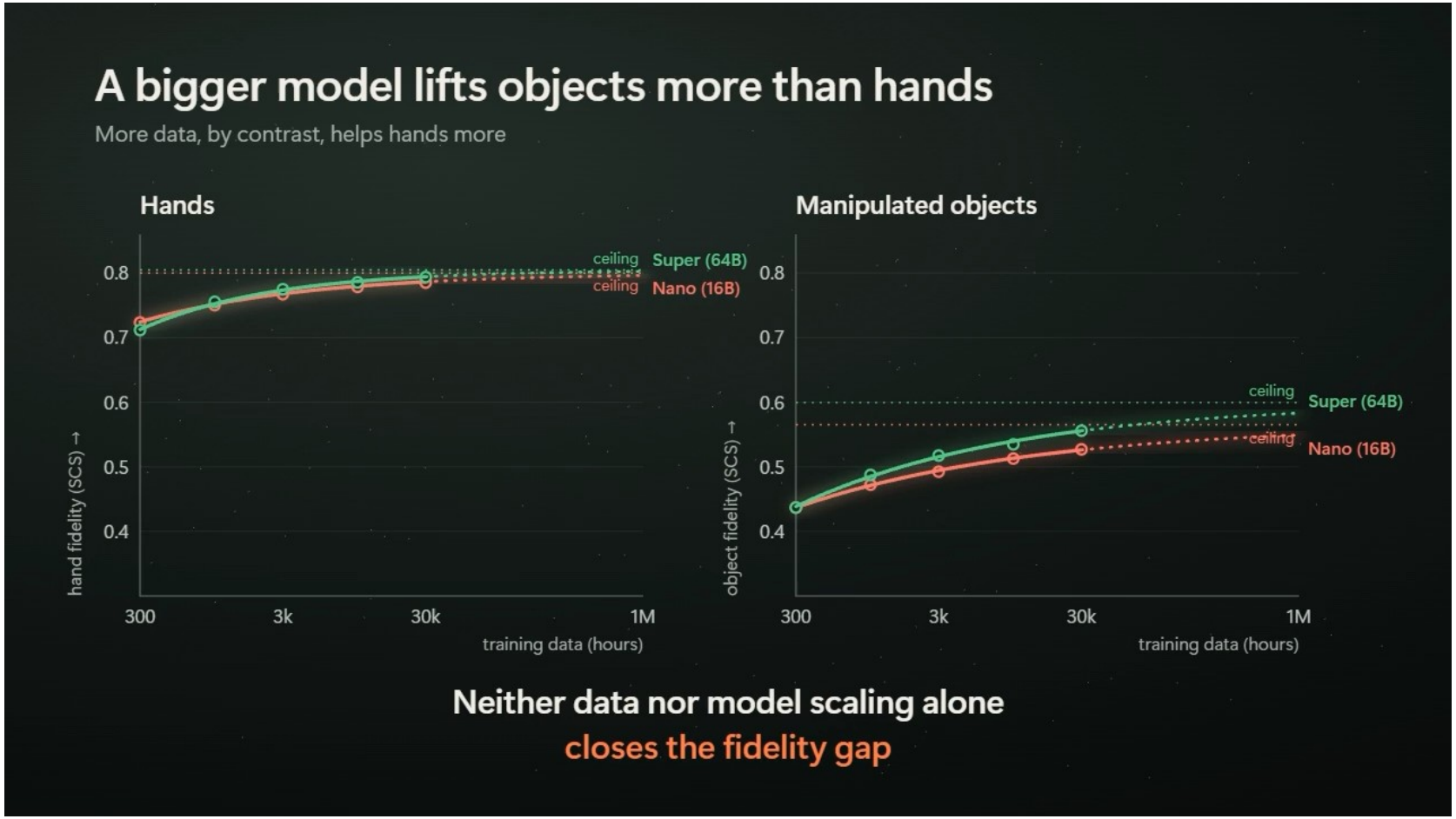
World Models from *Offline* Data

Scaling Data



World Models from *Offline* Data

Scaling Model Size



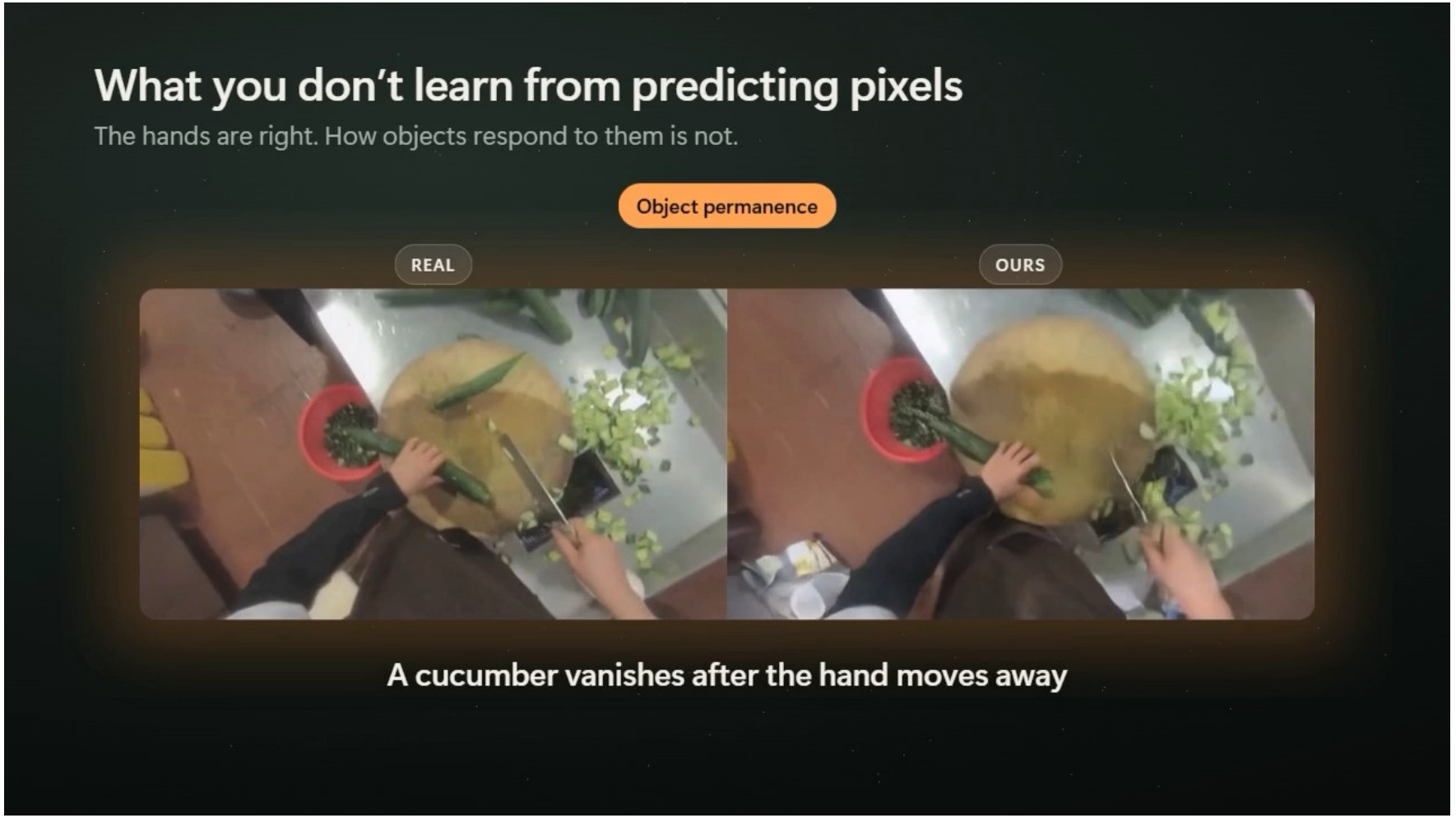
World Models from *Offline* Data

Can we raise the object ceiling? We change the training signal instead. We add extra noise where things move, so the model can't just copy pixels and has to actually predict their motion.



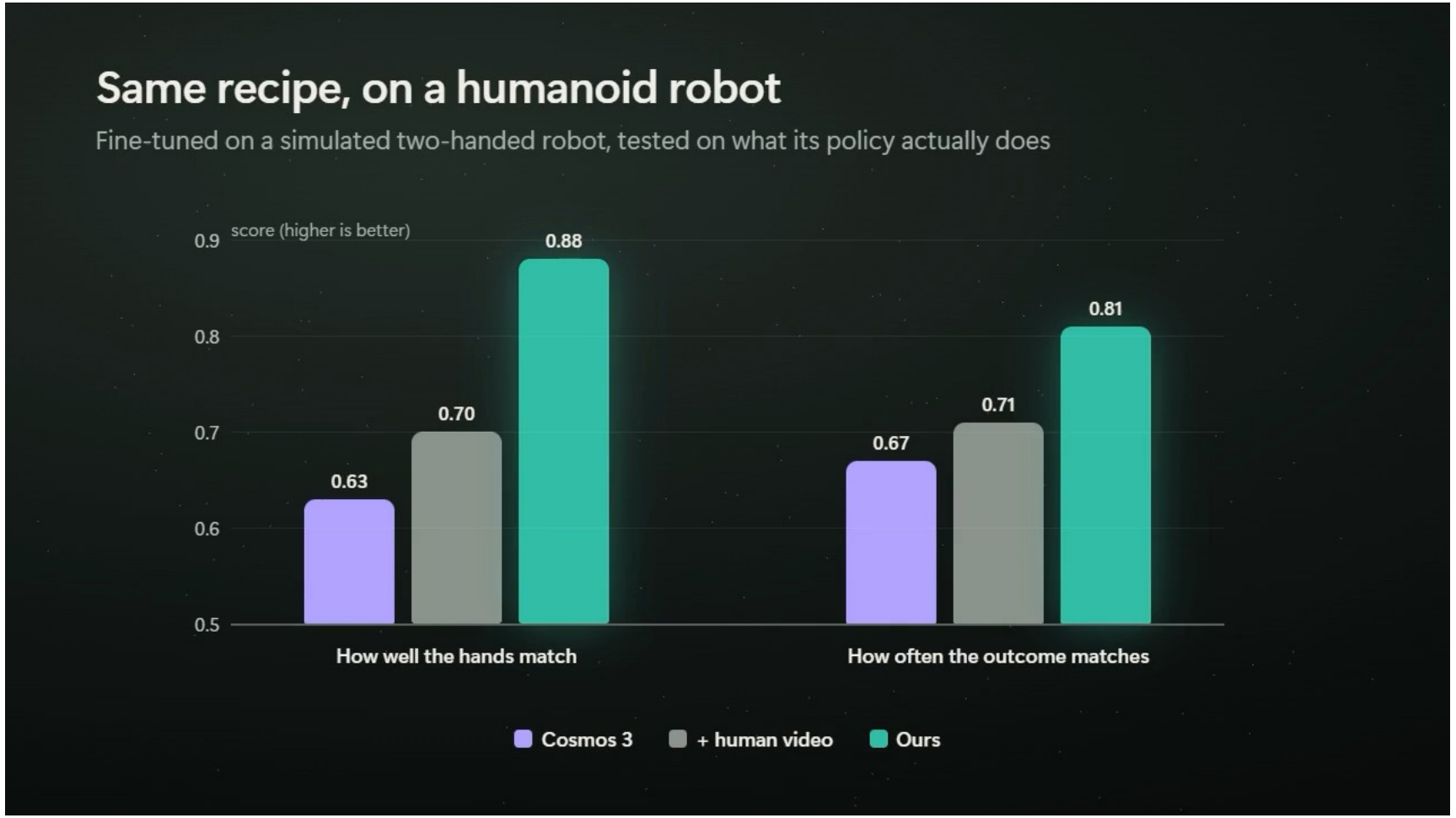
World Models from *Offline* Data

Issues with pixel predictions



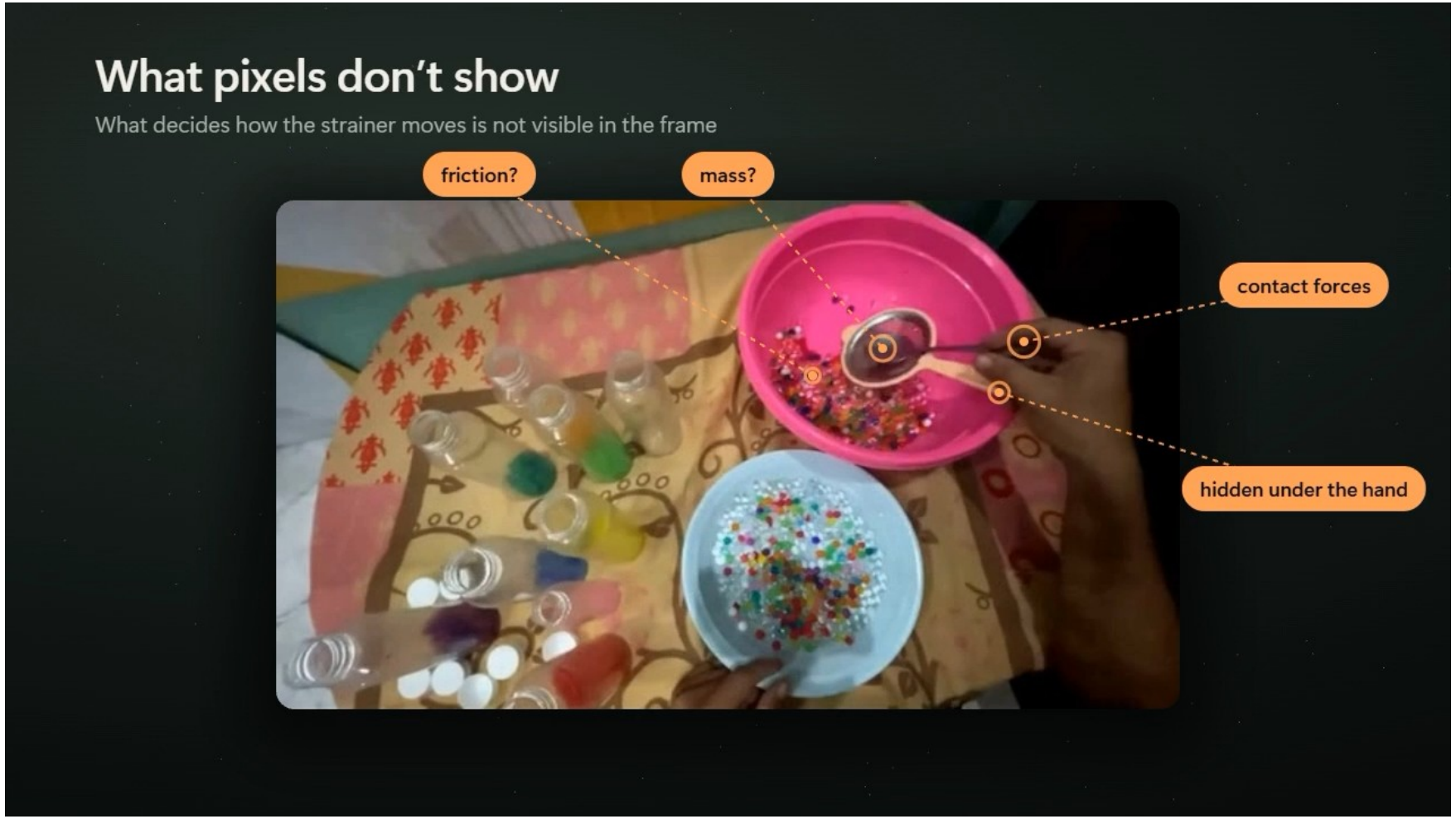
World Models from *Offline* Data

Analysis of fine-tuning on humanoid data



World Models from *Offline* Data

Takeaways



World Models from *Offline* Data

Beyond pixel predictions?

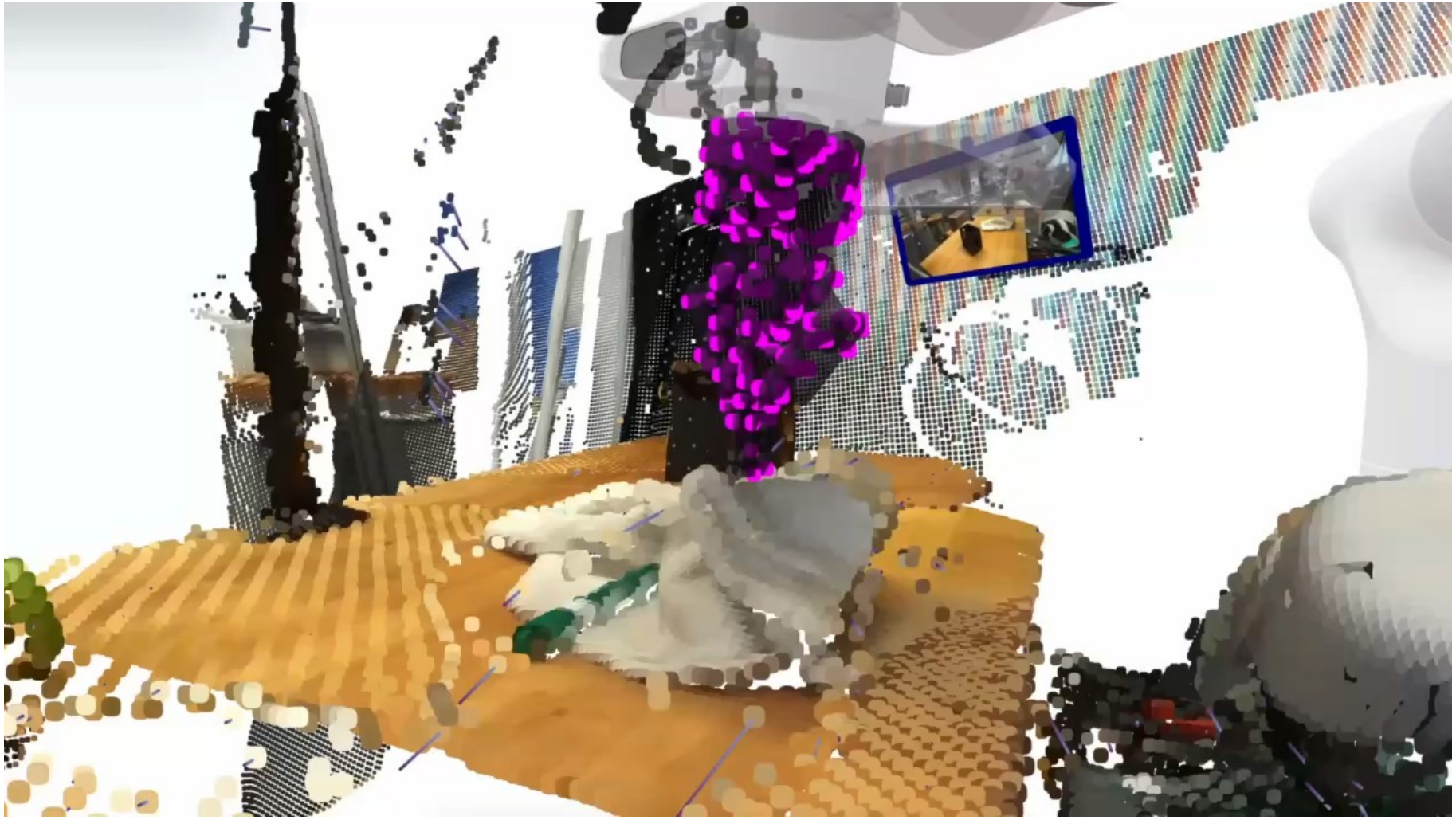
PointWorld: Scaling 3D World Models for In-The-Wild Robotic Manipulation

Wenlong Huang[1,†], Yu-Wei Chao[2], Arsalan Mousavian[2], Ming-Yu Liu[2], Dieter Fox[2], Kaichun Mo[2,*], Li Fei-Fei[1,*]

[1]Stanford University, [2]NVIDIA

[*] Equal advising | [†] Work done partly at NVIDIA

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Motion Tracks: A Unified Representation for Human-Robot Transfer in Few-Shot Imitation Learning

Juntao Ren[1], Priya Sundaresan[2], Dorsa Sadigh[2], Sanjiban Choudhury[1], Jeannette Bohg[2]
[1]Cornell University, [2]Stanford University

World Models from *Offline* Data

